

Entropy-Charged Soft Spectral Decompositions with Applications to Unique Games

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Abstract

We construct a soft spectral decomposition whose loss is charged to the information revealed about one root endpoint pair. For a half-lazy weighted graph, the leaf multipliers form a squared partition of unity, every leaf has low base threshold rank, and the total deleted edge mass is bounded by the component-conditional endpoint entropy. A rational late-time heat selection supplies the local energy-to-information bound; the entire tree has finite implicit access.

Combining this decomposition with the Guruswami–Sinop marginal-decoding framework gives an end-to-end Unique Games guarantee governed by the smaller of a vertexwise-information envelope and a component-entropy envelope. In the general worst case, an instance of optimum at least $1 - \kappa$ admits a labeling of error $O(\sqrt{\kappa \log(2/\alpha)/\alpha})$ in deterministic time $q^{O(n^\alpha)}$ times an input-bit polynomial. The dependence on κ remains a square root at fixed α .

We also compute the full split-channel information and prove a local logarithmic obstruction, together with the rank screen that prevents its cube example from lower-bounding this algorithm. Only proofs needed for these conclusions and their supporting refinements are included. The appendices give sharper ensemble and multiplier rounding, promise-free recovery, and secondary actual-loss-sensitive certificates.

1 Introduction

1.1 Overview and motivation

A Unique Games instance assigns a permutation constraint to each edge of a weighted graph. A labeling satisfies a constraint when its endpoint labels match under that permutation. For n vertices and an alphabet of size q , the near-perfect recovery problem asks for a labeling violating little constraint weight when some labeling violates at most a κ fraction of the total constraint weight. This is the algorithmic regime underlying Khot’s Unique Games Conjecture [1]. Our focus is the quantitative relationship between the promised error κ , the recovered error, and the running time.

Spectral algorithms exploit the fact that a highly satisfiable labeling has small constraint energy. On a graph with suitable expansion or low-dimensional spectral structure, this observation can be converted into an efficient recovery procedure. On an arbitrary graph, however, one must first find useful restrictions and account for everything lost between them. Arora–Barak–Steurer [2] established the subexponential-time decomposition approach, with a detailed near-perfect tradeoff in Steurer’s thesis [3]. This paper concentrates on the decomposition cost within that approach.

The guiding idea is to charge a split for the information it reveals, not to charge each recursion level independently. We use *soft* restrictions: a vertex can occur in several leaves, with nonnegative

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multipliers whose squares sum to one. These multipliers specify an actual retained mass for every edge. A single latent ordered endpoint pair then couples all levels of one decomposition tree, so the mutual-information chain rule bounds the total information spent. Component identification costs no edge mass and can be removed from this budget. The result is one principal structural contribution—an entropy-charged soft spectral decomposition—and a sequence of algorithmic consequences, rather than several unrelated algorithmic claims.

1.2 Principal results and their interpretation

A graph theorem independent of Unique Games. An endpoint law is a probability distribution a_{xy} on ordered vertex pairs. Here $a_{xy} = a_{yx}$, its marginal is $\mu(x) = \sum_y a_{xy} > 0$, and half-laziness means $a_{xx} \geq \mu(x)/2$. Its walk and scalar Dirichlet energy are $P(x, y) = a_{xy}/\mu(x)$ and $\mathcal{E}_P(f) = \frac{1}{2} \sum_{x,y} a_{xy} (f(x) - f(y))^2$. All logarithms and entropies use natural logarithms. Theorem 3.1 takes a rational symmetric half-lazy endpoint law a on at most n vertices, an integer $256 \leq R \leq n$, and rational parameters $0 < \nu \leq 1/n$ and $0 < \eta \leq 1/4$ for the minimum-mass threshold and spectral cutoff. Write $L = \log(1/\nu)$ and $u = L/\log R$. Under $3072\eta u \leq 1/34$, it constructs nonnegative rational leaf multipliers Φ_b with

$$\sum_b \Phi_b(x)^2 = 1, \quad Z_b = \sum_{x,y} a_{xy} \Phi_b(x) \Phi_b(y), \quad \text{rank } \mathbf{1}_{[0,\eta]}(I - P_b) \leq R^{5/8},$$

where P_b is the normalized walk of the corresponding weighted restriction. The rank counts, with multiplicity, its normalized Laplacian eigenvalues in the inclusive interval $[0, \eta]$. Its total deleted edge mass satisfies

$$D = 1 - \sum_b Z_b \leq 191488 \frac{\eta \log(2u)}{\log R} H(E \mid C) + 2n\nu.$$

Here E is the root ordered endpoint pair and C its connected component. Thus the entropy term concerns uncertainty *within* components, not the number or relative weights of the components alone. Any prescribed path is computable in $\text{poly}(n, \text{input length}, 1/\eta)$ bit operations. The associated sampler reaches leaf b with unconditional probability Z_b and otherwise dies. Exact rational sampling has expected polynomial bit cost; a bounded implementation permits an explicitly controlled additional failure probability. The construction does not explicitly list a polynomial-size family of leaves. Its overlapping restrictions are not a hard partition of the kind produced by the earlier ABS decomposition.

A complete-labeling consequence. For $n \geq 256$, $q \geq 2$, an integer $256 \leq R \leq n$, and rational $0 < \kappa < 1$, let OPT denote the maximum satisfied fraction of constraint weight. Under the promise $\text{OPT} \geq 1 - \kappa$, Theorem 4.2 combines the graph construction with a finite version of the Guruswami–Sinop conditional-marginal framework [4, 5]. Set

$$\nu = \frac{\kappa}{100n}, \quad u = \frac{\log(100n/\kappa)}{\log R}, \quad \alpha_H = \frac{\log(2u)}{2 \log R}.$$

The recovered error is $O(\sqrt{\kappa(u + \alpha_H H_*)})$. The static quantity H_* , defined in Section 4, is the minimum of two static information budgets obtained by integrating residual entropy envelopes. It is not a pointwise minimum taken before integration. One envelope uses the vertexwise distribution of endpoint surprisal, and therefore records where information is concentrated; the other uses the original component-conditional entropy. This is not merely a guarantee for one favorable leaf: it accounts for successive residual instances, interstage edges, and the final unprocessed mass. The

randomized implementation has time $q^{O(R^{5/8})} \text{poly}(n, q, b, 1/\kappa, \log(1/\xi))$ and success probability at least $1 - \xi$, for supplied rational $0 < \xi < 1$. The deterministic implementation has an additional explicit entropy-dependent path-search exponent. Here and below b denotes the binary input length, including supplied numerical parameters.

The uniform tradeoff and a promise-free implementation. Theorem 4.4 composes the preceding algorithm with the Gupta–Talwar logarithmic approximation [6]. This removes the numerical $1/\kappa$ cost and absorbs the deterministic path-search exponent in the general worst-case guarantee. In a common time-exponent parameter, Corollary 4.5 gives

$$\begin{aligned} \text{err}(\ell) &\leq \min \left\{ 1, 600\kappa \log(2n), C\sqrt{\frac{\kappa}{\alpha} \log \frac{2}{\alpha}} \right\}, \\ \text{time } &q^{O(n^\alpha)} \text{poly}(n, q, b), \quad 0 < \alpha < 5/8. \end{aligned} \tag{1}$$

Here α is supplied as a rational parameter. The constants are absolute, including for parameters varying with n . Corollary 4.6, stated in the main text and proved in Appendix B.1, removes the need to supply κ : its uniform bound is expressed in terms of the unknown error $1 - \text{OPT}$. Only $O(1 + \log \log n)$ localization calls are required. This wrapper does not assert that every instance-sensitive bound can likewise be made promise-free with the same parameters.

1.3 What is novel and what is derived

What is novel: the proposed structural contribution. The principal claim is the *specific loss–rank–information tradeoff* in Theorem 3.1, together with its finite soft-restriction construction. Its proof combines a late-time heat selection controlling energy per unit of support information (Lemma 3.2), a rational squared partition with quadratic loss (Lemma 3.3), and an exact posterior coupling that charges all smooth losses to $H(E \mid C)$. We do not claim to introduce spectral localization, heat evolution, or threshold rank; those belong to the preceding spectral and decomposition literature [7, 2, 3]. Nor is the mutual-information chain rule new [8, Chapter 2]. The proposed advance is making these ingredients yield the stated global loss bound for this explicitly defined soft object, without a branchwise entropy-monotonicity assumption.

What is derived: the algorithmic theorems. Theorem 4.2 is a consequence of that graph theorem and the Guruswami–Sinop tools [4, 5], with two application-specific arguments: transporting the conditional-marginal ensemble back through the multiplier, and controlling every residual from the immutable input. The vertexwise envelope in Lemma 4.1 makes that second step sensitive to the location, not just the total amount, of information. Theorem 4.4 and Corollaries 4.5 and 4.6 are further compositions with the classical fallback [6, 9] and finite parameter guessing. These are quantitative consequences of the principal construction, not claims of a new general SDP hierarchy or a new paradigm for rounding low-rank Unique Games.

What is inherited, and what the appendices refine. Column selection, the singleton embedding, and alphabet-efficient low-rank recovery are due to Guruswami–Sinop [4]; their local hierarchy framework [5] supplies the optimization architecture. Finite-precision ellipsoid optimization is classical [10]. Appendix A makes the required finite ensemble interface explicit and proves its soft transport. Its sharper coarea argument separates label disagreement from amplitude variation. Consider a good leaf with cutoff rank at most $R^{5/8}$ at $0 < \eta = 1024\bar{\kappa} \leq 1/4$, a labeling of error at

most $8\bar{\kappa}$, and $\mathcal{E}_P(\Phi_b)/Z_b \leq 8D$. Corollary A.5 produces a nonempty partial assignment with

$$\Gamma^2 \leq \frac{65}{432}\eta + \frac{2048}{27}D.$$

The quantity D is the loss of that invocation’s decomposition tree, and Γ is its partial assignment’s internal-violation-plus-twice-boundary mass, divided by assigned vertex mass, measured at the invocation root. Rank alone does not imply this bound. Direct multiplier thresholds are complementary candidates, with their identity-loop assumption and rank budget stated explicitly. These are supporting refinements of the transport interface, not separate general worst-case exponent claims. Appendix B supplies the weighted rational fallback implementation; its logarithmic approximation factor and tree-metric method remain those of [6, 9]. Appendix C groups secondary whole-tree loss estimates, guarded comparisons of fixed policies, and exactly certified flat partitions. The principal static entropy theorem does not depend on those optional procedures.

1.4 Related work and quantitative comparison

Polynomial-time recovery and structured graphs. Following Khot’s formulation [1], Trevisan [11] gave polynomial-time recovery of a constant fraction of constraints from sufficiently near-perfect instances with completeness error of order $1/\log n$. Gupta–Talwar [6] obtained the logarithmic error guarantee used here, through probabilistic tree embeddings [9]. Charikar–Makarychev–Makarychev [12] gave the alphabet-sensitive $O(\sqrt{\kappa \log q})$ bound, which is nontrivial when $\kappa \log q$ is sufficiently small. For weighted instances, the same per-edge estimate gives this expected-error bound by weighted Jensen’s inequality. These polynomial-time algorithms can be preferable to our general tradeoff, particularly for small alphabets.

Expansion gives a different source of favorable structure. Arora–Khot–Kolla–Steurer–Tulsiani–Vishnoi [13] established efficient recovery on expanding constraint graphs. Makarychev–Makarychev [14] sharpened this to error $O(\kappa/h_G)$ when $\kappa < c\lambda_G$, where h_G is edge expansion and λ_G the spectral gap in their normalization. Kolla [7] developed purely spectral recovery whose time depends on the relevant label-extended eigenspace. These works explain why spectral structure is useful, but assumptions about an expander or a label-extended eigenspace are not interchangeable with our theorem on arbitrary base graphs.

General decomposition and low-rank hierarchies. ABS [2] is the foundational general subexponential algorithm; Steurer’s thesis [3] supplies the explicit two-parameter near-perfect benchmark used below. The use of iterative partial assignments is already present there. Related divide-and-conquer algorithms were developed by Arora–Impagliazzo–Matthews–Steurer [15]. Barak–Raghavendra–Steurer [16] developed hierarchy rounding through global correlation, obtaining guarantees that recover the ABS constant-satisfaction subexponential regime and exploit low threshold rank on suitable instances. Guruswami–Sinop [4, Theorem 6.7 and Remark 6.8] give expected error at most

$$\left(1 + \frac{2 + \theta}{\lambda_{r+1}}\right) \kappa \quad \text{in time } q^{O(r/\theta)} \text{poly}(n),$$

Here $0 < \theta < 1$, the integer r satisfies $1 \leq r < n$, and $\lambda_{r+1} > 0$ is the $(r + 1)$ -st smallest eigenvalue of the base normalized Laplacian; a zero denominator gives no finite guarantee. Their Theorems 5.2 and 6.9 provide the column-selection and embedding ingredients used here. The local solver [5] is important: constructing a full hierarchy with an $n^{O(r)}$ cost would not give our stated alphabet-efficient interface. The error coefficient in our general tradeoff is attributed to the new decomposition analysis; alphabet-efficient decoding is explicitly inherited.

Result	Recovered-error bound	Running time and scope
Gupta–Talwar [6]	$O(\kappa \log n)$	Polynomial; general graphs.
Charikar–Makarychev– Makarychev [12]	$O(\sqrt{\kappa \log q})$	Polynomial; nontrivial for small $\kappa \log q$.
Steurer [3, Theorem 5.2]; ABS approach [2]	$O(\sqrt{\kappa/\alpha^3})$	$\exp(O(qn^\alpha))$; general graphs.
Guruswami–Sinop [4, 5]	$(1 + (2 + \theta)/\lambda_{r+1})\kappa$	$q^{O(r/\theta)} \text{poly}(n)$; spectral guarantee.
This paper, Corollary 4.5	$O(\sqrt{\kappa \log(2/\alpha)/\alpha})$	$q^{O(n^\alpha)} \text{poly}(n, q, b)$; general graphs.

Table 1: Relevant recovery guarantees, with errors capped by one. The displayed thesis benchmark has $\alpha \geq \log \log n / \log n$; our range is $0 < \alpha < 5/8$. Comparison of their error coefficients is on the common range. Polynomial factors are suppressed in the historical exponential-time entry. Randomized historical rounding bounds are stated in expectation; constant-success bounds follow with an absolute constant-factor loss. Alphabet dependence and spectral hypotheses differ.

Table 1 separates those hypotheses and time costs. At a common α , the stated error coefficient changes from $\alpha^{-3/2}$ in [3, Theorem 5.2] to $\sqrt{\log(2/\alpha)/\alpha}$ here. As α tends to zero, the ratio of the former coefficient to the latter is $1/(\alpha\sqrt{\log(2/\alpha)})$. This is a comparison of proved upper-bound expressions, not a lower bound on an optimized earlier algorithm. Nor are the alphabet costs identical: $\exp(O(qn^\alpha))$ and $q^{O(n^\alpha)}$ have different q dependence; for fixed q both have the form $\exp(O(n^\alpha))$ up to polynomial factors. At fixed α , both displayed bounds retain the $\sqrt{\kappa}$ completeness power. The improvements claimed here are therefore in the decomposition-sensitive coefficient and the instance-sensitive information accounting, not in that power or in every parameter regime.

Other relevant algorithms and analytic tools. Hopkins–Schramm–Trevisan [17] obtain a subexponential Unique Games gap-distinguishing guarantee using Sherali–Adams linear programs, without SDP rounding or subspace enumeration. This is an alternative algorithmic framework; the gap statement alone is not the same two-parameter near-perfect recovery guarantee as (1). Alev–Lau [18] use low-diameter decompositions for minor-free and bounded-genus constraint graphs. For structured *affine* Unique Games, Bafna–Barak–Kothari–Schramm–Steurer [19] develop rounding on certified small-set expanders, and Bafna–Minzer [20] treat globally hypercontractive graphs. These graph-class and constraint-type assumptions matter; our comparison does not claim dominance over their specialized guarantees. The conditioned shift potential in [19, Definition 3.4] concerns weighted collision probabilities of relative labels in a sum-of-squares pseudodistribution. Our budget is instead the Shannon entropy of an actual root endpoint pair, conditioned on its component, and controls recursive deletion.

The analytic ingredients also have established antecedents. Heat evolution and small-support Rayleigh quotients are central to spectral profile methods [21]; the entropy chain rule and relative-entropy identities are standard [8]; and the cube analysis in Section 5 uses classical hypercontractivity [22]. We prove the quantitative combinations needed here rather than assigning novelty to these tools. This discussion covers the directly relevant algorithmic and analytic lines of work, not an exhaustive survey or certification of priority. Independent specialist correctness and priority review remain necessary.

1.5 Overview of the proof strategy

1. Find an informative low-energy split. At a node with all vertex masses at least ν and base cutoff rank exceeding $R^{5/8}$, the lazy walk $Q = (I + P)/2$ has $\text{Tr}(Q^{2T}) > \sqrt{R}$ at the time used in Lemma 3.2. Select a mean-one density column g_t with squared norm $N_T = \|g_T\|_\mu^2 > \sqrt{R}$. For $F_t = \log \|g_t\|_\mu^2 > 0$ and $e_t = \mathcal{E}_P(g_t) / \|g_t\|_\mu^2$, heat contraction gives $e_t \leq 2(F_t - F_{t+1})$. Monotonicity of F_t then yields

$$\sum_{t < T} \frac{e_t}{F_t} \leq 2 \log(F_0/F_T) \leq 2 \log(2u).$$

Selecting a late time and truncating the density turns this estimate into an energy-to-support-information bound containing $\log(2u)$. A stereographic parametrization gives a squared partition of unity with quadratic energy loss. A fixed dyadic mesh preserves the bound while discarding the heat-vector denominators before recursion; the encoding length of accumulated multipliers still grows linearly with path depth. These are the local estimates in Lemmas 3.2 and 3.3.

2. Charge the entire tree to one information budget. Draw one root endpoint pair $E = (x, y)$. At a split, select child i with probability $\chi_i(x)\chi_i(y)$ and otherwise kill the path. Nonnegativity and the squared partition of unity make these probabilities valid by Cauchy–Schwarz. Conditional on reaching a node, Bayes’ rule gives exactly that node’s weighted endpoint law. Thus the chain rule controls the reach-weighted sum of local information by the root entropy, even though entropy need not decrease along an individual surviving branch. Route connected components first to replace $H(E)$ by $H(E | C)$. Combining this budget with the local loss-per-information estimate gives Theorem 3.1; cleanup contributes the explicit $2n\nu$.

3. Decode a good leaf without reconciling overlapping leaves. At an admitted recovery invocation, the unconditional path law gives a constant probability of reaching a leaf with both a good restricted witness and small multiplier energy *relative to retained leaf mass*, as stated in Lemma 3.4. The finite conditional-marginal construction uses the inherited Guruswami–Sinop ingredients [4, 5]. Winner-gap rounding creates a one-hot amplitude field. Multiplication by the leaf multiplier and ensemble coarea then produce a nonempty partial assignment, selected by its actual rational violation-plus-boundary charge. Keeping label disagreement separate from amplitude variation sharpens this interface. Only one accepted partial assignment is committed at a time, so no agreement between separately rounded overlapping leaves is required.

4. Preserve the information gain through complete recovery. After each accepted partial assignment, rebuild the residual from the immutable original instance and replace outgoing mass by identity loops. Lemma 4.1 bounds the resulting entropies through the original vertexwise profile and component information. Integrating the partial-assignment charges over remaining mass proves Theorem 4.2, including its stopping and terminal charges. The classical fallback handles parameter ranges in which localization would have excessive numerical or deterministic search cost. Exact satisfiability testing and a bounded geometric guessing schedule then give the promise-free corollary, including termination on false guesses.

5. Delimit what the current analysis does not show. Section 5 computes the full child-or-kill information and proves a logarithmic obstruction on the broad high-trace class. Its cube witness already satisfies the low-rank stopping criterion, however, so the algorithm does not split it. That witness is therefore not a lower bound for nodes the algorithm actually splits. The hypercontractive

estimate is classical [22]; the channel calculation and the rank-screen qualification specify the limitation of this argument. Neither a log-free global loss bound nor a better general completeness power follows from this analysis.

1.6 Organization of the paper

Section 2 fixes the weighted input, normalization, constraint-energy, and soft-restriction conventions. Section 3 proves the standalone spectral decomposition and its finite access guarantees. Section 4 establishes the residual information envelopes, complete recovery, uniform tradeoff, and promise-free statement. Section 5 gives the actual-channel analysis and precisely scoped limitation. Appendix A proves the required finite decoder and its sharper transport, coarea, and direct-multiplier refinements. Appendix B gives the classical fallback and the proof of the promise-free wrapper. Appendix C contains the optional actual-loss-sensitive extension. It can be omitted on a first reading without interrupting the principal decomposition–recovery–limitation narrative.

2 Notation and normalization

Definition 2.1 (Unique Games input). An instance has $n \geq 1$ vertices, alphabet $[q] = \{1, \dots, q\}$ with integer $q \geq 1$, and finitely many records (u, v, π, w) , where π is a permutation of $[q]$ and $w \geq 0$ is rational. A labeling $\ell : V \rightarrow [q]$ satisfies a record when $\ell(v) = \pi(\ell(u))$. If the total weight is positive, its error $\text{err}(\ell)$ is the weight of violated records divided by the total weight, and $\text{OPT} = 1 - \min_{\ell} \text{err}(\ell)$. For zero total weight we set every labeling’s error to zero and $\text{OPT} = 1$. Loops and parallel records are allowed. The input length includes the explicit permutations and all rational weights and parameters. Running times below count bit operations.

Preprocessing and orientation. Discard all zero-weight records before any constraint check or graph construction. If none remain, return arbitrary labels and perform no normalization. All subsequent normalized-law statements concern the positive-total-weight case. Vertices incident to no remaining record are likewise omitted from all laws and operators and receive arbitrary output labels; n remains the original vertex-count bound. All later records refer to this preprocessed instance or to its explicitly specified transforms.

Normalize the positive record weights to sum to one. Replace a record (u, v, π, p) by $(u, v, \pi, p/2)$ and $(v, u, \pi^{-1}, p/2)$, even when $u = v$; these are record occurrences, so coincident endpoints do not remove either contribution. The two satisfaction tests are equivalent to the original one. Write b_{xy} for their aggregate endpoint mass and $\mu^0(x) = \sum_y b_{xy} > 0$ for its marginal. Reversal makes b symmetric, of total mass one, without changing any labeling’s error.

Half-lazification and immutable residuals. Retain half the directed record mass and add the identity-permutation loop of mass $\mu^0(x)/2$ at each vertex. The scalar endpoint law is

$$a_{xy}^0 = \frac{1}{2}b_{xy} + \frac{1}{2}\mathbf{1}_{x=y}\mu^0(x).$$

It is symmetric, has marginal μ^0 , and is half-lazy: $a_{xx}^0 \geq \mu^0(x)/2$. The new loops are always satisfied, so the error in this normalized record law is exactly $\text{err}(\ell)/2$. We retain constraint records, including nonidentity loops, for decoding; only their scalar endpoint masses are aggregated for entropy and base-graph spectral calculations.

For a residual vertex set U with $t = \mu^0(U) > 0$, start again from these immutable half-lazified records. Keep the internal records, replace each record from $x \in U$ to $y \notin U$ by an identity loop at

x of the same mass, discard records whose source is outside U , and divide all retained masses by t . Its scalar endpoint law is

$$a_{xy}^U = \frac{a_{xy}^0 + \mathbf{1}_{x=y} \sum_{z \notin U} a_{xz}^0}{t} \quad (x, y \in U).$$

Summing a row gives marginal $\mu_U(x) = \mu^0(x)/t$, hence total mass one. Off-diagonal symmetry is preserved, and the retained original identity-loop mass alone is at least $\mu_U(x)/2$. An original witness of error at most κ has residual error at most $\kappa/(2t)$, since completion adds no violations and internal violated mass is at most $\kappa/2$; later we may use the weaker bound κ/t .

Let $E_U = (X_U, Y_U)$ be a random ordered pair with law a^U , and let C_U be its connected-component variable in the scalar graph with edges of positive mass. Put $H_U = H(E_U)$ and $h_U = H(E_U | C_U)$. The original pair and its component variable are denoted $E^0 \sim a^0$ and C_0 , respectively. Equivalently, to sample E_U , sample an endpoint pair from a^0 conditional on its first endpoint lying in U , then move the second endpoint to the first when it lies outside U . These completed residuals are the recovery invocation roots; soft leaf weights are never carried over to the next invocation.

Scalar and constraint energies. Let a be a nonnegative symmetric scalar law of total mass one on its positive-degree vertices. Write $\mu(x) = \sum_y a_{xy} > 0$, $P_{xy} = a_{xy}/\mu(x)$, and, for real scalar fields f, g ,

$$\langle f, g \rangle_\mu = \sum_x \mu(x) f(x) g(x), \quad \mathcal{E}_P(f) = \frac{1}{2} \sum_{x,y} a_{xy} (f(x) - f(y))^2 = \langle f, (I - P)f \rangle_\mu.$$

The identity follows by expansion and symmetry. In particular P is a self-adjoint Markov operator in $L^2(\mu)$. If $D = \text{diag}(\mu)$, its symmetric normalized Laplacian is

$$H = I - D^{-1/2} a D^{-1/2} = D^{1/2} (I - P) D^{-1/2}.$$

Thus H in Euclidean coordinates and $I - P$ in $L^2(\mu)$ have the same cutoff eigenvalues and multiplicities. All cutoff ranks in this paper count the inclusive interval $[0, \eta]$ unless stated otherwise.

When the law carries constraints, a_e denotes the mass of a directed record $e : u \rightarrow v$, so $\sum_e a_e = 1$ and $a_{uv} = \sum_{e: u \rightarrow v} a_e$. Every record has its equally weighted reverse with permutation π_e^{-1} ; an identity loop may be its own reverse. For a vector $z \in \mathbb{R}^q$, define the pullback by $(T_e z)_i = z_{\pi_e(i)}$. Fiber fields $z_u \in \mathbb{R}^q$ use the degree-weighted norm $\|z\|_\mu^2 = \sum_u \mu(u) \|z_u\|_2^2$ and energy

$$\hat{\mathcal{E}}_P(z) = \frac{1}{2} \sum_e a_e \|z_u - T_e z_v\|_2^2.$$

We omit the subscript when the law is specified. A one-hot field $z_u = e_{\ell(u)}$, with e_i the i th standard basis vector, has norm one. Its squared difference at a record is zero when that record is satisfied and two otherwise; hence its energy is exactly the labeling error in the current normalized record law. At the initial half-lazified root this is half the original input error.

The permutation lift has a directed edge of mass a_e from (u, i) to $(v, \pi_e(i))$ for every e and $i \in [q]$. Reverse records make this lift symmetric. Its degree measure is $\hat{\mu}(u, i) = \mu(u)$, of total mass q , not the probability measure $\mu(u)/q$. Consequently its scalar Dirichlet form, with this unnormalized degree measure, is exactly $\hat{\mathcal{E}}_P$; no factor of q is introduced. A loop at u with a nonidentity permutation can join different lift coordinates, so it can have positive constraint energy even though its scalar base-graph energy is always zero.

Soft restrictions and their degree measure. Here a is the half-lazy scalar law at the fixed root of one decomposition invocation, with positive marginal μ . For a nonzero nonnegative multiplier Φ on this vertex set, define

$$M_\Phi = \sum_u \mu(u)\Phi(u)^2, \quad Z_\Phi = \sum_{u,v} a_{uv}\Phi(u)\Phi(v), \quad M_\Phi - Z_\Phi = \mathcal{E}_P(\Phi).$$

The energy identity again follows by expansion. The raw restricted degree and its normalized marginal are

$$d_\Phi(x) = \sum_y a_{xy}\Phi(x)\Phi(y), \quad \mu_\Phi(x) = d_\Phi(x)/Z_\Phi.$$

These normalizations are well-defined on $\text{supp } \Phi$, because half-laziness and symmetry give

$$d_\Phi(x) \geq \frac{1}{2}\mu(x)\Phi(x)^2, \quad 0 < M_\Phi/2 \leq Z_\Phi \leq M_\Phi.$$

For the upper bound, sum $2\Phi(x)\Phi(y) \leq \Phi(x)^2 + \Phi(y)^2$ against a_{xy} . The restricted scalar law is $a_{xy}^\Phi = a_{xy}\Phi(x)\Phi(y)/Z_\Phi$, and its constraint-record masses, when present, are $a_e\Phi(u)\Phi(v)/Z_\Phi$. Vertices with zero multiplier are deleted. In general μ_Φ is not $\mu\Phi^2/M_\Phi$.

For example, on two vertices take $a_{xy} = 1/4$ for every pair and $\Phi = (1/2, 1)$. Then $M_\Phi = 5/8$, $Z_\Phi = 9/16$, and $d_\Phi = (3/16, 3/8)$, giving $\mu_\Phi = (1/3, 2/3)$, whereas $\mu\Phi^2/M_\Phi = (1/5, 4/5)$. The restricted walk has $P_{11}^\Phi = 1/3 < 1/2$, although the root was half-lazy.

All accumulated multipliers and their masses above refer to the fixed invocation root; local heat and split computations instead use the current normalized node law. A Unique Games recovery root has identity-loop mass at least half its degree. Descendants need not retain that pointwise fraction relative to their new degrees, as the example illustrates even for scalar diagonal mass. At every node $Q = (I + P)/2$ is used for analytic laziness: reversibility and the spectrum of a Markov operator give $0 \preceq Q \preceq I$. Its scalar Dirichlet form satisfies $\mathcal{E}_Q(f) = \mathcal{E}_P(f)/2$. Using this heat operator does not apply a second half-lazification to the constraint objective.

Entropy conventions. All logarithms are natural unless a base is displayed. For a finite random variable with probabilities p_z , entropy is $-\sum_{z:p_z>0} p_z \log p_z$; zero-probability summands are omitted also in conditional entropy and mutual information. Endpoint entropy concerns the scalar ordered pair, not the record identity, permutation, or labels. Since an edge never joins distinct connected components, its component variable can be read from either endpoint.

3 Entropy-charged spectral decomposition

The construction below is a graph theorem: it does not use a labeling promise or an alphabet. Its output is an implicitly accessible family of overlapping weighted restrictions, not a hard partition.

Theorem 3.1 (Soft spectral decomposition). *Let a be a rational symmetric endpoint law on at most n vertices, with positive marginal μ and $a_{xx} \geq \mu(x)/2$. Fix an integer $256 \leq R \leq n$ and rational $0 < \nu \leq 1/n$, $0 < \eta \leq 1/4$. Put*

$$L = \log(1/\nu), \quad u = L/\log R, \quad J = \log(2u), \quad 3072\eta u \leq 1/34.$$

There is a deterministic finite binary tree whose internal nodes carry nonnegative rational local multipliers χ_0, χ_1 satisfying $\chi_0^2 + \chi_1^2 = 1$, with nonempty proper child supports. Its leaf products Φ_b , extended by zero to the root, satisfy

$$\sum_b \Phi_b(x)^2 = 1, \quad M_b = \sum_x \mu(x)\Phi_b(x)^2, \quad Z_b = \sum_{x,y} a_{xy}\Phi_b(x)\Phi_b(y), \quad M_b/2 \leq Z_b \leq M_b.$$

Each normalized restriction $a_{xy}^b = a_{xy}\Phi_b(x)\Phi_b(y)/Z_b$ has positive degrees and

$$\text{rank } \mathbf{1}_{[0,\eta]}(I - P_b) \leq R^{5/8}. \quad (2)$$

The cutoff is inclusive and the projector is taken in the weighted Hilbert space. If $E \sim a$, and C is its root connected-component variable, write $h = H(E \mid C)$. Then

$$D := \sum_b \mathcal{E}_P(\Phi_b) = 1 - \sum_b Z_b \leq \min \left\{ 1 - \sum_x a_{xx}, 191488 \frac{\eta J}{\log R} h + 2n\nu \right\}. \quad (3)$$

In particular $D \leq 1/2$, $\sum_b M_b = 1$, and $h \leq 2 \log n$.

The tree has depth at most $n - 1$. Any specified path can be computed in $\text{poly}(n, b_{\text{in}}, 1/\eta)$ bit operations, with absolute polynomial degree; b_{in} includes the encoding of ν and η . An exact sampler, with expected polynomial bit cost, reaches leaf b with unconditional probability Z_b and otherwise dies. For any integer $q \geq 1$, a capped version has at most 2^{-q} additional failure probability and bit cost polynomial also in q . For the exact sampler, let X count smooth child-1 choices before the sampler reaches a leaf or dies; then $\mathbb{E}X \leq 4h/\log R$. Write $X(b)$ for this count along the path to leaf b . For every integer $k \geq 0$, every leaf with $X(b) \leq k$ can be found deterministically in at most

$$n \sum_{j=0}^{\min\{k, n-1\}} \binom{n-1}{j} \quad (4)$$

path simulations. Full-tree enumeration is not claimed to be polynomial.

Here and below, H and mutual information use natural logarithms. We prove the theorem through two local estimates and one global information calculation.

3.1 A rational choice of heat time

Lemma 3.2. Suppose a current node has a rational normalized symmetric endpoint law, all marginal masses at least ν , and rank in (2) exceeding $R^{5/8}$. With the parameters of Theorem 3.1, a rational computation finds a nonzero rational vector $h_1 \geq 0$ and rational λ such that

$$\sigma := \mu(\text{supp } h_1) \leq R^{-1/4}, \quad \mathcal{E}_P(h_1) \leq \lambda \|h_1\|_\mu^2, \quad \eta \leq \lambda \leq 3072\eta u, \quad \frac{\lambda}{\log(1/\sigma)} \leq \frac{44J}{T},$$

where $T = \lfloor \log_2 R \rfloor / (32\eta)$ and $\log R / (128\eta) \leq T \leq \log R / (16\eta)$.

Proof. Let $Q = (I + P)/2$. It is positive semidefinite in $L^2(\mu)$ even when the current node is not half-lazy. Here $u \geq 1$, so $\eta \leq 1/104448$. Writing $\ell = \lfloor \log_2 R \rfloor \geq 8$, we have $\ell \geq \frac{1}{2} \log_2 R$ and $\ell/(64\eta) \leq T \leq \ell/(32\eta)$; these imply $T \geq 2$ and the stated bounds. Every eigenvalue of $I - P$ at most η contributes at least $(1 - \eta/2)^{2T} \geq \exp(-2\eta T) \geq R^{-1/8}$ to $\text{Tr}(Q^{2T})$, where $\log(1 - \eta/2) \geq -\eta$ for $0 \leq \eta \leq 1/4$. Thus rank failure implies $\text{Tr}(Q^{2T}) > \sqrt{R}$.

For density columns $g_{t,i} = Q^t \mathbf{1}_i / \mu(i)$, reversibility gives $\sum_i \mu(i) \|g_{t,i}\|_\mu^2 = \text{Tr}(Q^{2T})$. Choose a column maximizing $N_T = \|g_T\|_\mu^2$. Every g_t is nonnegative with mean one, $N_0 = 1/\mu(i) \leq 1/\nu$, and $N_T > \sqrt{R}$. The spectral expansion writes N_t as a nonnegative combination of q^{2t} , $0 \leq q \leq 1$, so N_t is nonincreasing and log-convex by Cauchy-Schwarz (with $q^0 = 1$ also for $q = 0$). Set $F_t = \log N_t$ and $e_t = \mathcal{E}_P(g_t)/N_t$. Since $I - Q^2 \succeq I - Q$,

$$F_0 \leq L, \quad F_t \geq F_T > \frac{1}{2} \log R \geq \log 16, \quad e_t \leq 2(1 - N_{t+1}/N_t) \leq 2(F_t - F_{t+1}).$$

For positive $F_t \geq F_{t+1}$, the last difference divided by F_t is at most $\log(F_t/F_{t+1})$. Consequently

$$\sum_{t=0}^{T-1} e_t/F_t \leq 2\log(F_0/F_T) \leq 2J. \quad (5)$$

Among $\lfloor T/2 \rfloor \leq t < T$, some time has $e_t/F_t \leq 4J/T$. At each candidate time form $h_t = (g_t - N_t/4)_+$. Markov's inequality gives $\sigma_t \leq 4/N_t \leq R^{-1/4}$, while $\|\min(g_t, N_t/4)\|_\mu^2 \leq N_t/4$ follows from $\min(g_t, N_t/4)^2 \leq (N_t/4)g_t$ pointwise. The reverse triangle inequality now gives $\|h_t\|_\mu \geq \sqrt{N_t} - \sqrt{N_t}/2$, so $h_t \neq 0$ and $\|h_t\|_\mu^2 \geq N_t/4$. The scalar positive-part map is 1-Lipschitz and therefore decreases energy. Hence

$$\frac{\mathcal{E}_P(h_t)}{\|h_t\|_\mu^2} \leq 4e_t, \quad \log(1/\sigma_t) \geq F_t/2.$$

For clarity, log-convexity makes $d_t = F_t - F_{t+1}$ nonnegative and nonincreasing. Thus $td_t \leq \sum_{s < t} d_s = F_0 - F_t$, and every candidate $t \geq T/3$ satisfies $e_t \leq 2(F_0 - F_t)/t \leq 6L/T$.

No transcendental comparison is used to select a time. Compute $\lambda_t = \max\{\eta, \mathcal{E}_P(h_t)/\|h_t\|_\mu^2\}$ and $m_t = \lfloor \log_2 N_t \rfloor$ by rational power comparisons, and minimize λ_t/m_t . Here $m_t \geq 4$ and $m_t \log 2 \leq F_t < \frac{5}{4}m_t \log 2$. Also $\eta/F_t \leq 1/(8T) \leq J/(4T)$. The good time from (5) therefore has $\lambda_t/F_t \leq 17J/T$; rational selection loses at most $5/4$. Indeed, for a minimizing time s and the good time t , $\lambda_s/F_s \leq \lambda_s/(m_s \log 2) \leq \lambda_t/(m_t \log 2) < (5/4)\lambda_t/F_t$. The chosen pair obeys $\lambda/F_s < 22J/T$ and hence the claimed ratio. Finally $\eta T \leq \log R/16 \leq L/16$, so at every candidate $\lambda_t \leq 24L/T \leq 3072\eta u$. All vectors are rational matrix powers; their bit bounds are supplied in the construction below. $\square$

3.2 A squared partition with quadratic loss

Lemma 3.3. *Let the current rational node and h_1, λ, σ satisfy the hypotheses and conclusion of Lemma 3.2; in particular, every current marginal mass is at least ν . One can compute nonnegative rational multipliers with proper nonempty supports and $\chi_0^2 + \chi_1^2 = 1$ such that, in the current normalized law,*

$$p_i = \sum_{x,y} a_{xy} \chi_i(x) \chi_i(y), \quad K = 1 - p_0 - p_1 \leq 34\lambda p_1, \quad \mu(\text{supp } \chi_1) \leq \sigma.$$

Every multiplier coordinate has $O(\log(1/\eta) + \log(1/\nu))$ bits, independently of the heat vector's denominator.

Proof. For $r = h_1/\max h_1$, first use $\chi_1 = 2r/(1+r^2)$ and $\chi_0 = (1-r^2)/(1+r^2)$. Writing $F(r) = ((1-r^2)/(1+r^2), 2r/(1+r^2))$, we have $\|F(r) - F(s)\|_2^2 = 4(r-s)^2/((1+r^2)(1+s^2))$ on $[0, 1]$. In particular F is 2-Lipschitz, and $\chi_1 \geq r$. Thus, with $M_1 = \|\chi_1\|_\mu^2$,

$$K = \mathcal{E}_P(\chi_0) + \mathcal{E}_P(\chi_1) \leq 4\mathcal{E}_P(r) \leq 4\lambda M_1.$$

Choose the largest dyadic mesh $\Delta = 2^{-j}$ satisfying $\Delta^2 \leq \eta\nu/10000$, round r down to this mesh, and apply the same map, denoting the results by primes. The vector perturbation $v = F(r') - F(r)$ is pointwise at most 2Δ . Reversibility gives $\mathcal{E}_P(v) \leq 2\|v\|_\mu^2 \leq 8\Delta^2$, with vector energy defined as the sum of its coordinate energies. The squared triangle inequality in edge space and vertex space gives

$$K' \leq 2K + 16\Delta^2, \quad M_1 \leq 2M_1' + 8\Delta^2.$$

A maximum remains equal to one, so the minimum-mass hypothesis gives $M'_1 \geq \nu$. As $\eta \leq \lambda \leq 1/34$,

$$K' \leq 16\lambda M'_1 + (64\lambda + 16)\Delta^2 \leq 17\lambda M'_1.$$

For the last inequality use $(64\lambda + 16)\Delta^2 \leq 18\eta\nu/10000 \leq \lambda M'_1$. Since $p'_1 = M'_1 - \mathcal{E}_P(\chi'_1) \geq (1 - 17\lambda)M'_1$, we have $p'_1 \geq M'_1/2$, giving $K' \leq 34\lambda p'_1$. The nonzero vector h_1 has proper support because $\sigma \leq R^{-1/4} \leq 1/4$. Hence r attains both zero and one. Rounding retains these values and cannot enlarge the small support; both child supports therefore remain nonempty and proper. The dyadic numerators have $O(j)$ bits and the displayed rational map has the same bound. Drop the primes. $\square$

3.3 Construction, entropy accounting, and finite access

Proof of Theorem 3.1. Fix vertex order and every tie-breaking rule. Apply the first applicable rule at each current node:

1. At a singleton, stop.
2. If disconnected, split off the first connected component using its indicator and complementary indicator, and recurse on both children.
3. Otherwise compute the inclusive rank r , and stop if $r^8 \leq R^5$.
4. If the rank test fails and a current marginal mass is below ν , isolate its first such vertex by an indicator split (“cleanup”), and recurse on both children.
5. Otherwise use Lemmas 3.2–3.3 and recurse on both children. Call this a smooth split.

Every tree edge strictly reduces the nonempty support, so the depth is at most $n - 1$. Each leaf satisfies the rank bound: a singleton has rank one, and every other leaf passes the exact rank test. In particular, cleanup ensures the minimum-mass hypothesis whenever a smooth split is performed.

For any accumulated multiplier Φ , persistent root diagonal mass gives

$$d_\Phi(x) = \sum_y a_{xy} \Phi(x) \Phi(y) \geq \frac{1}{2} \mu(x) \Phi(x)^2.$$

Here Φ is the product along a node’s path, extended by zero outside its support; the root product is 1. The upper bound $Z_\Phi \leq M_\Phi$ follows by applying $2\Phi(x)\Phi(y) \leq \Phi(x)^2 + \Phi(y)^2$ and summing. Thus $Z_\Phi > 0$, $M_\Phi/2 \leq Z_\Phi \leq M_\Phi$, and all retained degrees are positive. The normalized node marginal is d_Φ/Z_Φ , not $\mu\Phi^2/M_\Phi$. Induction on the finite tree gives the squared partition of unity. Index internal nodes by their binary prefixes t . If $w_t = Z_{\Phi_t}$ is a node’s root mass, then $w_{ti} = w_t p_{t,i}$ and $w_t - w_{t0} - w_{t1} = w_t K_t$. Therefore

$$\sum_b M_b = 1, \quad D = 1 - \sum_b Z_b = \sum_t w_t K_t = \sum_b \mathcal{E}_P(\Phi_b). \quad (6)$$

Here is the information calculation with its exact posterior law. Draw one latent endpoint pair $E = (x, y)$ from the root. At a reached node choose child i with conditional probability $\chi_{t,i}(x)\chi_{t,i}(y)$ and die with the remaining probability. These probabilities are nonnegative and sum to at most one by Cauchy–Schwarz. The conditional probability of reaching a prefix is $\Phi_t(x)\Phi_t(y)$; hence its unconditional reach probability is w_t and its endpoint posterior is exactly the node law. Let I_t be mutual information between this posterior endpoint and the full child-or-death outcome. Let $\mathcal{T}$ be

the finite transcript padded after termination, and let $\mathcal{R}_0$ consist of the initial indicator-routing nodes that separate the root connected components. If the root is connected, $\mathcal{R}_0$ is empty. Routing has no death and gives a deterministic prefix code for C ; after it ends, every subsequent node belongs to a single root component. In particular C is determined by $\mathcal{T}$, even if the path later dies. The chain rule gives $\sum_t w_t I_t = I(E; \mathcal{T})$ and $\sum_{t \in \mathcal{R}_0} w_t I_t = H(C)$. Subtracting yields

$$\sum_{t \notin \mathcal{R}_0} w_t I_t = I(E; \mathcal{T} \mid C) \leq H(E \mid C) = h. \quad (7)$$

Here and in the preceding sum, only internal nodes are included. The initial routing contribution $H(C)$ is not charged to h ; later component cuts may consume conditional information but cost no mass. At a smooth node, outcome 1 forces both endpoints into its support, whose prior probability is at most its first-endpoint mass σ_t . The relative-entropy support bound yields $I_t \geq p_{t,1} \log(1/\sigma_t)$: indeed for a distribution supported on an event of prior probability z , its divergence from the prior equals its divergence from the conditioned prior plus $\log(1/z)$. The other outcome contributions are nonnegative. Thus

$$\sum_{t \text{ smooth}} w_t K_t \leq \frac{1496J}{T} \sum_{t \text{ smooth}} w_t I_t \leq 191488 \frac{\eta J}{\log R} h, \quad \mathbb{E}X = \sum_{t \text{ smooth}} w_t p_{t,1} \leq \frac{4h}{\log R}.$$

The equality for $\mathbb{E}X$ counts a smooth child-1 transition even on a path that subsequently dies. Cleanup at x costs $K_t = 2 \sum_{y \neq x} a_{xy}^t < 2\nu$. The reach-weighted number of such steps is at most the maximum depth, so their total cost is at most $2n\nu$. This proves the entropy bound; $h \leq H(E) \leq \log(n^2)$ because there are at most n^2 ordered endpoint pairs. Finally $k(x, y) = \sum_b \Phi_b(x) \Phi_b(y)$ lies in $[0, 1]$ and has $k(x, x) = 1$. Hence $D = \sum_{x,y} a_{xy} (1 - k(x, y))$ is at most $1 - \sum_x a_{xx} \leq 1/2$.

For completeness, the construction is rational in the bit model. At a node with rational symmetric raw matrix A and positive degree matrix D , count the nonpositive eigenvalues of $(1 - \eta)D - A$. Congruence by $D^{-1/2}$ identifies this count with the inclusive normalized cutoff rank. Rational symmetric elimination uses a nonzero diagonal as a one-dimensional pivot; if all remaining diagonals vanish, a nonzero off-diagonal supplies an invertible two-dimensional pivot of negative determinant. An all-zero remainder contributes only zero eigenvalues. The signs of the one-dimensional pivots and the inertia $(1, 1)$ of each two-dimensional pivot determine the count, including all zeros. To bound bit length, clear the input denominators once. Each Schur complement entry is a quotient of determinants of submatrices of the permuted original integer matrix, whose pivot block is nonsingular. The row-norm determinant bound gives polynomial bit lengths for these integers. Reducing rational entries after each operation therefore keeps all pivot computations polynomial in the node's encoding length. Thus neither an exact irrational eigenbasis nor a numerical equality test is needed.

All accumulated multipliers have $O(nj)$ bits per coordinate by the precision reset; $j = O(\log(1/\eta) + \log(1/\nu))$ is fixed by the root parameters. Compute raw node weights directly from the root law and these products. Raw weights, their sums, and normalized degrees then have polynomial encoding length, regardless of how small Z_Φ is numerically. If b_Q is a common integer denominator for the current rational Q , each entry of Q^t has denominator dividing b_Q^t ; numerators have $O(t \log b_Q)$ bits up to polynomial dimension terms. Since $T = O(\log R/\eta)$, the heat computation and its rational score comparisons have the promised cost. Only the rounded multipliers are propagated; the heat denominators are discarded. The support-depth bound gives the same polynomial cost for a path.

Computational sampling uses only the rational probabilities p_0, p_1, K ; the latent pair is not sampled or needed. Express these probabilities with a common integer denominator B . For $B > 1$, draw $\lceil \log_2 B \rceil$ fair bits, accept the resulting integer if it is below B , and otherwise repeat; $B = 1$ is

deterministic. The accepted integer is uniform on $\{0, \dots, B-1\}$, and the acceptance probability is greater than $1/2$. Assigning consecutive intervals of the appropriate integer lengths gives an exact draw in expected polynomial bit cost. There are at most $n-1$ draws on a path. Capping each draw at $s = q + \lceil \log_2 n \rceil$ attempts adds failure probability at most $n2^{-s} \leq 2^{-q}$ and gives a polynomial bit budget. This is an additional failure event coupled to the exact sampler, not a renormalization of the surviving leaf law. To obtain (4), fix any vertex x in a leaf support and record the absolute depths at which its path takes smooth child 1. Depths are numbered $0, \dots, n-2$. At every indicator split, including cleanup, follow the unique child containing x ; at a smooth split follow the recorded choice. Abort if x leaves the selected support. The recorded pair reconstructs its leaf exactly. Enumerating vertices and depth subsets gives the asserted count and covers every required leaf. $\square$

Lemma 3.4 (A good leaf under the unconditional sampling law). *If the root carries a Unique Games witness of error at most $\bar{\kappa} \geq 0$ and $D \leq 1/4$, the exact sampler of Theorem 3.1 reaches with probability at least $1/2$ a leaf on which that witness has error at most $8\bar{\kappa}$ and $\mathcal{E}_P(\Phi_b)/Z_b \leq 8D$.*

Proof. Let v_b be the root-law mass of violated constraint records weighted by $\Phi_b(x)\Phi_b(y)$. The witness error in the normalized leaf is v_b/Z_b . Since $k(x, y) \leq 1$ recordwise, $\sum_b v_b \leq \bar{\kappa}$. Also $\sum_b Z_b(\mathcal{E}_P(\Phi_b)/Z_b) = D$ by (6). Under the unconditional leaf law each bad event has probability at most $1/8$ by Markov's inequality. Death costs at most $1/4$, leaving probability at least $1/2$. When either bound is zero its nonnegative summands vanish, so the same conclusion holds without division by that bound. $\square$

The theorem also gives an exact deleted-edge interpretation. Put $a_{xy}^{\text{loss}} = a_{xy}(1 - k(x, y))$ and $d^{\text{loss}}(x) = \sum_y a_{xy}^{\text{loss}}$. This is a nonnegative symmetric measure of total mass D . For any such measure m , use the unnormalized quadratic form

$$\mathcal{E}_m(f) := \frac{1}{2} \sum_{x, y} m_{xy} (f(x) - f(y))^2.$$

In particular $\mathcal{E}_{a^{\text{loss}}}$ involves no division by D and is zero when $D = 0$. For every real-valued f on the root vertices, restricting f to a leaf support in terms involving P_b , we have

$$\begin{aligned} \mathcal{E}_P(f) &= \sum_b Z_b \mathcal{E}_{P_b}(f) + \mathcal{E}_{a^{\text{loss}}}(f), \\ \sum_b \mathcal{E}_P(\Phi_b f) &= \sum_b Z_b \mathcal{E}_{P_b}(f) + \sum_x d^{\text{loss}}(x) f(x)^2 \leq \mathcal{E}_P(f) + D \|f\|_\infty^2. \end{aligned}$$

The first identity follows from $a_{xy} = a_{xy}k(x, y) + a_{xy}^{\text{loss}}$. For the second, expanding its left side gives $\sum_x \mu(x) f(x)^2 - \sum_{x, y} a_{xy}k(x, y) f(x) f(y)$; the sum of leaf energies has the same cross term and diagonal coefficient $\sum_y a_{xy}k(x, y) = \mu(x) - d^{\text{loss}}(x)$. Finally $\mathcal{E}_{a^{\text{loss}}}(f) \geq 0$ and $\sum_x d^{\text{loss}}(x) = D$ give the inequality. Thus D measures actual omitted edge mass, not an auxiliary entropy potential alone.

4 Entropy-sensitive recovery of a complete labeling

4.1 The immutable information envelope

Throughout this section $n \geq 256$ and $q \geq 2$. As in Section 2, zero-total-weight inputs are returned immediately with arbitrary labels; the normalized-law definitions and profile-dependent

guarantees below concern positive-total-weight inputs. Put $E^0 \sim a^0$, $M = 2 \log n$, $H_0 = H(E^0)$ and $h_0 = H(E^0 \mid C_0)$. For every positive-degree vertex define

$$r_x = \frac{1}{\mu^0(x)} \sum_{y: a_{xy}^0 > 0} a_{xy}^0 \log \frac{1}{a_{xy}^0}, \quad \mathcal{T}(t) = \max_{\substack{0 \leq z_x \leq 1 \\ \sum_x \mu^0(x) z_x = t}} \sum_x \mu^0(x) r_x z_x \quad (0 \leq t \leq 1).$$

The fractional relaxation defines the profile at every mass, not only at masses attained by actual vertex subsets. For $0 < s, t \leq 1$, with $h_{\text{bin}}(s) = -s \log s - (1-s) \log(1-s)$ and zero terms omitted, define

$$\begin{aligned} g(s) &= \min \{M, \max \{0, \min \{\mathcal{T}(s)/s + \log s, (H_0 - h_{\text{bin}}(s))/s\}\}\}, \\ G(t) &= \sup_{t \leq s \leq 1} g(s), \quad H_G = \frac{1}{4} \left(\int_0^1 \sqrt{G(t)/t} dt \right)^2, \\ \psi(\theta) &= \theta \left(1 + \frac{1}{2} \log(1/\theta)\right)^2 \quad (0 < \theta \leq 1), \quad \psi(0) = 0, \\ H_* &= \min \{H_G, M\psi(h_0/M)\}. \end{aligned}$$

The quantities r_x include vertex surprisal, not merely transition-row entropy. The supremum defining G is needed because g need not be decreasing. The integral defining H_G is improper at zero; it converges because $0 \leq G(t) \leq M$ and $t^{-1/2}$ is integrable. No value of $g(0)$ is needed. These functions describe the guarantee and are not oracles used by the algorithm.

Lemma 4.1 (Residual information bounds). *The profile $\mathcal{T}$ is continuous, nondecreasing and concave, with endpoints $0, H_0$, and $\mathcal{T}(t)/t$ is nonincreasing on $(0, 1]$. Every nonempty completed residual of mass t satisfies*

$$tH_U \leq \mathcal{T}(t) + t \log t, \quad tH_U \leq H_0 - h_{\text{bin}}(t), \quad th_U \leq h_0.$$

Consequently

$$h_U \leq H_U \leq G(t), \quad H_0 \leq G(t) \leq \min \{M, \mathcal{T}(t)/t\}, \quad h_U \leq \min \{M, h_0/t\}.$$

In particular $H_0 \leq H_G \leq M$ and $h_0 \leq M\psi(h_0/M) \leq M$.

Proof. Ordering vertices by decreasing r_x fills the maximizing fractional set in that order: moving mass from a lower value to an unsaturated higher value cannot decrease its objective. Thus $\mathcal{T}$ is piecewise linear with nonnegative decreasing slopes. Concavity and $\mathcal{T}(0) = 0$ imply that $\mathcal{T}(t)/t$ decreases and is at least H_0 .

Condition $E^0 = (X, Y)$ on $X \in U$ and map (X, Y) to (X, X) when $Y \notin U$. This deterministic map has output law E_U . Its input entropy is

$$H(E^0 \mid X \in U) = t^{-1} \sum_{x \in U} \mu^0(x) r_x + \log t.$$

Entropy decreases under a deterministic map, and the indicator of U is feasible in the fractional maximization, proving the first inequality. For $0 < t < 1$, the chain rule gives

$$H_0 = h_{\text{bin}}(t) + tH(E^0 \mid X \in U) + (1-t)H(E^0 \mid X \notin U).$$

Discard the last nonnegative term to prove the second inequality; $t = 1$ follows directly from the deterministic map.

Let original components have masses π_j , and let $t_j = \mu^0(U \cap C_j)$. Applying the same argument within component j gives, when $t_j > 0$,

$$t_j H(E_U \mid C_0 = j) \leq \pi_j H(E^0 \mid C_0 = j).$$

Here the event has conditional probability t_j/π_j in component j , so the same chain rule justifies the factor π_j . In the residual, component j has probability t_j/t . Summing the displayed inequality gives $tH(E_U \mid C_0) \leq h_0$. Boundary completion creates no edges between original components, so C_U refines C_0 and further conditioning only decreases entropy. This proves $th_U \leq h_0$.

At a feasible residual mass both expressions inside the inner minimum in g bound $H_U \geq 0$; also $H_U \leq M$. Hence $H_U \leq g(t) \leq G(t)$. At every mass, $g(s) \leq \min\{M, \mathcal{T}(s)/s\}$, so taking the supremum over $s \geq t$ proves the asserted upper bound for G . Nested supremum domains make G nonincreasing, while $g(1) = H_0$ gives its lower bound. Integrating $\sqrt{H_0/t}$ and $\sqrt{M/t}$ proves the bounds on H_G . Finally $\theta \leq \psi(\theta) \leq 1$: $\psi(1) = 1$, its limit at zero is zero, and $\psi'(\theta) = \frac{1}{2} \log(1/\theta) + \frac{1}{4}(\log(1/\theta))^2 \geq 0$. $\square$

4.2 The end-to-end theorem

Theorem 4.2 (Entropy-sensitive complete recovery). *Let $n \geq 256$, $q \geq 2$, and $256 \leq R \leq n$ be integers, and let $0 < \kappa < 1$ be rational. Given an instance with $\text{OPT} \geq 1 - \kappa$, put*

$$L = \log(100n/\kappa), \quad u = L/\log R, \quad \alpha_H = \frac{\log(2u)}{2\log R}, \quad \mathcal{B}_* = u + \alpha_H H_*.$$

There is an absolute C and a component-first algorithm producing, on success, a complete labeling of error at most

$$\min\{1, C\sqrt{\kappa\mathcal{B}_*}\}.$$

For rational $0 < \xi < 1$, it has a bounded randomized implementation with success probability at least $1 - \xi$ and time

$$q^{O(R^{5/8})} \text{poly}(n, q, \text{input length}, 1/\kappa, \log(1/\xi)).$$

It also has a deterministic implementation of time

$$\exp\left(O\left(R^{5/8} \log q + \frac{\mathcal{H}_* \log n}{\log R}\right)\right) \text{poly}(n, q, \text{input length}, 1/\kappa),$$

where

$$\mathcal{H}_* = \begin{cases} \min\{G(\sqrt{\kappa u}), h_0/\sqrt{\kappa u}\}, & \kappa u < 1, \\ 0, & \kappa u \geq 1. \end{cases}$$

One may take $C = 2^{23}$. No decision requires evaluating G, H_G or an improper integral.

We first give the finite decision quantities used in its proof. For a rational probability vector p put

$$\bar{H}(p) = \sum_{p_z > 0} p_z \left(\frac{7}{10} k_z + \frac{7}{5} (1 - 2^{k_z} p_z) \right), \quad k_z = \lfloor \log_2(1/p_z) \rfloor.$$

For current component masses σ_j and their normalized laws b^j , set

$$\bar{h}_U = \min\{\bar{H}(E_U), \sum_j \sigma_j \bar{H}(b^j)\}.$$

Define the global rational numbers

$$b = \lceil \log_2(100n/\kappa) \rceil, \quad s = \lfloor \log_2 R \rfloor, \quad U_0 = b/s, \quad j = \lceil \log_2(2U_0) \rceil, \quad \bar{J} = 7j/10, \quad \bar{d} = 100/(69s),$$

and the stage budgets

$$B_U = u + \alpha_H h_U, \quad \bar{B}_U = U_0 + \frac{1}{2} \bar{d} \bar{J} \bar{h}_U.$$

Lemma 4.3 (Exact rational budget). *The quantities with overbars have polynomial bit complexity and*

$$h_U \leq \bar{h}_U \leq \frac{7}{5} h_U, \quad u \leq B_U \leq \bar{B}_U < 4B_U, \quad u \leq U_0 \leq \frac{81}{64} u.$$

The entropy estimates preserve exact zero.

Proof. For $x \in [1/2, 1]$, the tangent and chord bounds give $1 - x \leq -\log x \leq 2 \log 2(1 - x)$. Since $69/100 < \log 2 < 7/10$, writing $-\log p_z = k_z \log 2 - \log(2^{k_z} p_z)$ yields $H(p) \leq \bar{H}(p) \leq (7/5)H(p)$. Both terms defining $\bar{h}_U$ majorize h_U , and its second term is at most $(7/5)h_U$. Binary comparisons compute every integer exponent; their magnitudes and all rational encodings are polynomially bounded.

Put $a = \log_2(100n/\kappa)$ and $c = \log_2 R$. Then $a \geq c \geq 8$, $a \leq b < a + 1$, and $s \leq c < s + 1$. Thus $U_0/u = (b/a)(c/s)$ lies in $[1, (9/8)^2]$. These bounds and $69/100 < \log 2 < 7/10$ also give

$$\frac{1}{\log R} \leq \bar{d} < \frac{105}{92 \log R}, \quad \log(2u) \leq \bar{J} < \frac{5}{2} \log(2u).$$

Indeed $\bar{d} \log R < 100(9/8)(7/10)/69 = 105/92$. For the last upper bound use $\log(81/64) < 1/4$ and $j < \log_2(2U_0) + 1$ to obtain $\bar{J} < (70/69) \log(2u) + 1$, and then $\log(2u) > 69/100$. The entropy coefficient inflates by less than $(105/92)(7/5)(5/2) = 735/184 < 4$; the u coefficient inflates by at most $81/64$. This proves the budget comparison. $\square$

Proof of Theorem 4.2. Decisions and existence of accepted candidates. Let ρ be the lesser of one and the least power of two that is at least $\sqrt{\kappa U_0}$. Thus, analytically, $\rho = \min\{1, 2^{\lceil \log_2 \sqrt{\kappa U_0} \rceil}\}$. Comparisons of rational squares implement this rule without computing the square root or logarithm. Power-of-two rounding loses less than a factor two, and $\sqrt{81/64} = 9/8$, giving

$$\rho < 3\sqrt{\kappa u}, \quad \rho \geq \sqrt{\kappa u} \quad \text{if } \rho < 1.$$

At residual mass t , stop and label the residual arbitrarily if $t \leq \rho$, or if $(\kappa/t)\bar{B}_U > 2^{-34}$. Otherwise invoke the component-first tree with

$$\bar{\kappa} = \kappa/t, \quad \eta = 1024\bar{\kappa}, \quad \nu = \kappa/(100n).$$

The restricted witness has error at most $\bar{\kappa}$; this is a statement about the complete mixture, not each component separately. Admission and $\bar{B}_U \geq u \geq 1$, $t \leq 1$ give $\kappa \leq \bar{\kappa} \leq 2^{-34}$ and $\eta B_U \leq 2^{-24}$, in particular $\eta \leq 2^{-24} < 1/4$. Hence $3072\eta u \leq 3 \cdot 2^{-14} < 1/34$. Theorem 3.1 therefore applies and gives

$$D \leq 3 \cdot 2^{17} \eta B_U + \kappa/50 \leq 3/128 + 2^{-34}/50 < 1/32.$$

Here $191488\eta \log(2u)h_U / \log R = 382976\eta \alpha_H h_U \leq 3 \cdot 2^{17} \eta B_U$, and $2n\nu = \kappa/50$. By the good-leaf and finite-decoder interfaces (Lemma 3.4 and Corollary A.5), a sampled path reaches a good leaf with unconditional probability at least $1/2$. On every such leaf the bounded decoder, supplied

with the invocation-root records, the leaf product Φ_b , and $\kappa_0 = 8\bar{\kappa}$, enumerates a nonempty partial labeling S whose actual invocation-root charge obeys

$$\Gamma = \frac{B + 2C_{\text{out}}}{\mu_U(S)}, \quad \Gamma^2 \leq \frac{65}{432}\eta + \frac{2048}{27}D < 2^{38}\bar{\kappa}B_U.$$

Here B is violated internal directed mass and C_{out} is outgoing directed mass, both in the normalized invocation root. The decoder hypothesis $128\kappa_0 \leq \eta$ holds with equality; the good-leaf event supplies its witness-error and loss bounds. Test the exactly rational condition $\Gamma^2 \leq 2^{38}\bar{\kappa}\bar{B}_U$ on every enumerated candidate; accept one that passes, install it and remove its vertices. Decoder failure at an inspected leaf simply continues the search. At least one candidate passes on every good leaf. There are at most n successful invocations, and every accepted score satisfies $\Gamma \leq 2^{20}\sqrt{\kappa B_U/t}$.

One proof for both entropy envelopes. Apply the following argument separately to

$$W(t) = G(t), \quad H_W = H_G, \quad \text{and} \quad W(t) = \min\{M, h_0/t\}, \quad H_W = M\psi(h_0/M).$$

In either case W is nonnegative and nonincreasing, $h_U \leq W(t)$, and

$$\int_0^1 \sqrt{W(t)/t} dt = 2\sqrt{H_W}.$$

For the second case this follows by splitting at h_0/M : the integral is $\sqrt{h_0}(2 + \log(M/h_0))$, with value zero when $h_0 = 0$. Let $F(t) = u + \alpha_H W(t)$. If accepted stage i removes immutable mass $\Delta_i = t_i - t_{i+1}$, then $B_{U_i} \leq F(t_i)$ and $\sqrt{F(t)/t}$ is nonincreasing in t . Consequently $\Delta_i \sqrt{F(t_i)/t_i} \leq \int_{t_{i+1}}^{t_i} \sqrt{F(t)/t} dt$. Summing disjoint stage intervals and using $\sqrt{u + \alpha_H W} \leq \sqrt{u} + \sqrt{\alpha_H W}$ gives

$$\sum_i \Delta_i \Gamma_i \leq 2^{20}\sqrt{\kappa} \int_0^1 \sqrt{F(t)/t} dt \leq 2^{21}\sqrt{2\kappa(u + \alpha_H H_W)}.$$

The last inequality also uses $\sqrt{u} + \sqrt{\alpha_H H_W} \leq \sqrt{2(u + \alpha_H H_W)}$. This holds for every accepted trajectory, not merely in expectation.

The terminal mass t_f must also be charged; if all vertices were removed, put $t_f = 0$. A tail stop gives $t_f < 3\sqrt{\kappa u}$. At a budget stop, $t_f < 2^{36}\kappa B_{U_f}$, and therefore

$$t_f^2 < 2^{36}\kappa(ut_f + \alpha_H t_f h_{U_f}) \leq 2^{36}\kappa(u + \alpha_H H_W).$$

For the component envelope use $t_f h_{U_f} \leq h_0 \leq H_W$; for G use $t_f h_{U_f} \leq t_f H_{U_f} \leq H_0 \leq H_G$. For the charging identity, if stage i removes S_i , then $\mu_{U_i}(S_i) = \Delta_i/t_i$ and $\Delta_i \Gamma_i = t_i(B_i + 2C_{\text{out},i})$. This pays original half-lazified internal violations at that stage and both orientations of every edge to a later stage. Completion loops are identities and contribute no violation; edges to earlier stages have already been paid. Only terminal internal records remain unpaid, with total mass at most t_f . Thus, undoing half-lazification,

$$\text{err}_{\text{original}} \leq \min\{1, 2[\sum_i \Delta_i \Gamma_i + t_f]\} \leq \min\{1, 2^{23}\sqrt{\kappa(u + \alpha_H H_W)}\}.$$

The terminal bounds give $t_f \leq 2^{18}\sqrt{\kappa(u + \alpha_H H_W)}$, and $2(2^{21}\sqrt{2} + 2^{18}) < 2^{23}$ verifies the displayed constant. This includes zero-entropy and trivial-bound cases. Both envelopes analyze the same decisions and success event, so their minimum gives H_* . We have not replaced it by the integral of a new combined pointwise profile.

Bounded random search. At each invocation use $N_{\text{path}} = \lceil \log_2(2n/\xi) \rceil$ fresh killed paths (Lemma 3.4). Conditional on any earlier successful history, failure to find a good leaf has probability at most $2^{-N_{\text{path}}}$. A union bound over at most n invocations costs at most $\xi/2$. There are at most $n^2 N_{\text{path}}$ rational branch draws. To sample a rational categorical law, draw a uniform integer with the required denominator by binary rejection, whose acceptance probability is at least $1/2$. Cap each draw at $\lceil \log_2(2n^2 N_{\text{path}}/\xi) \rceil$ trials. Coupling with uncapped draws bounds the additional failure probability by $\xi/2$. A cap failure, or exhaustion of the path batch without an accepted candidate, is reported as failure and the remaining labels are arbitrary. No exact conditional leaf law is claimed after capping.

Deterministic search. At an invoked root let $K_U = \lceil 16\bar{d}\bar{h}_U \rceil$. The sampler satisfies $\mathbb{E}X \leq 4h_U/\log R$, where X counts smooth small-child choices. Markov's inequality gives $\Pr(X > K_U) \leq 1/4$; when $h_U = 0$, both K_U and X vanish. Thus some good leaf has at most K_U counted choices.

The support-vertex encoding in (4) covers every such leaf with $\exp(O((K_U + 1)\log n))$ path simulations; it does not enumerate hard component and cleanup decisions independently. If no candidate passes after this finite enumeration, report failure and label the remaining vertices arbitrarily. This exit cannot occur under the promised witness; specifying it also bounds runs on false guesses. The rational estimates give $K_U < 26h_U/\log R + 2$. Every invoked root has $t > \rho \geq \sqrt{\kappa u}$, so Lemma 4.1 gives $h_U \leq \mathcal{H}_*$. If $\kappa u \geq 1$, the tail test invokes no root and the zero cap is valid.

Finally every visited tree node has polynomial bit cost in n , the input length and $1/\kappa$, by finite access in Theorem 3.1; the original reset bounds apply equally after zero-one component restrictions. The decoder costs $q^{O(R^{5/8})}$ times that polynomial. Rational residual rebuilding, entropy estimates and charge comparisons are polynomial. Multiplying by the stated path counts and at most n invocations proves the two running times. $\square$

4.3 The general accuracy–time tradeoff

The entropy-sensitive implementation retains numerical $1/\kappa$ work and a deterministic path-search term. The next composition is a separate worst-case guarantee; it does not remove those costs from Theorem 4.2 with its sharper instance-specific envelope.

Theorem 4.4 (Completeness-uniform worst-case guarantee). *Let $n \geq 256$, $q \geq 2$, and $256 \leq R \leq n$ be integers, and let $0 < \kappa < 1$ be rational. For an instance with $\text{OPT} \geq 1 - \kappa$, put*

$$v = \frac{\log n}{\log R}, \quad B_n = v(1 + \log(2v)).$$

A deterministic algorithm produces a labeling with

$$\text{err}(\ell) \leq \min\{1, C_{\text{GT}}\kappa \log(2n), C'\sqrt{\kappa B_n}\}$$

in time $q^{O(R^{5/8})} \text{poly}(n, q, \text{input length})$, where C_{GT} and C' are absolute. The polynomial has no numerical $1/\kappa$ factor.

Proof. Always run the deterministic Gupta–Talwar fallback (Lemma B.2), whose error is at most $C_{\text{GT}}\kappa \log(2n)$ and whose time is polynomial in the binary input length. Set $l = \lceil \log_2 n \rceil$. Also run the deterministic algorithm of Theorem 4.2 precisely when $R > l^4$ and $n\kappa \geq 1$. Return the labeling with least actual rational error. All branch tests are rational or integer comparisons.

When that invocation is made, $1/\kappa \leq n$ and $n \geq 256 > 100$, so $L \leq \log(100n^2) \leq 3 \log n$, whence $u \leq 3v$. Since $H_* \leq 2 \log n$,

$$B_* \leq u + v \log(2u) \leq 3v + v \log(6v) \leq 3v(1 + \log(2v)) = 3B_n.$$

The last inequality uses $\log 3 \leq 2 \log(2v)$ for $v \geq 1$. Also $R^{5/8} > l^{5/2} \geq (\log n)^2$, which absorbs the additional deterministic path exponent, since $\mathcal{H}_* \leq 2 \log n$ and $\log q \geq \log 2$. The numerical cost is polynomial in n because $1/\kappa \leq n$ on this branch.

If $\kappa < 1/n$, the fallback alone suffices:

$$\kappa \log(2n) \leq \sqrt{\kappa} \frac{\log(2n)}{\sqrt{n}} \leq \sqrt{\kappa} \leq \sqrt{\kappa B_n}.$$

Here $\log(2n) \leq \sqrt{n}$ for $n \geq 256$, as follows at 256 and thereafter by differentiating $\sqrt{n} - \log(2n)$.

It remains to handle $R \leq l^4$. Write $x = \log n$, $y = \log R$. Since $l \leq x/\log 2 + 1 \leq 2x$ for $n \geq 256$, we have $y \leq 4 \log(2x)$ and $\log y \leq 3y/16 + 1$: the maximum of $\log y - 3y/16$ is $\log(16/3) - 1 < 1$. Hence

$$B_n = \frac{x}{y}(1 + \log(2x) - \log y) \geq x/16.$$

Every labeling has error at most one, and $\min\{1, z\} \leq \sqrt{z}$ for $z \geq 0$. The fallback therefore has error at most

$$\sqrt{C_{\text{GT}} \kappa \log(2n)} \leq \sqrt{32 C_{\text{GT}} \kappa B_n}.$$

Taking $C' = \max\{\sqrt{3}C, C_{\text{GT}}, \sqrt{32 C_{\text{GT}}}\}$ handles all branches. Its polynomial fallback cost and the preceding large- R calculation prove the claimed uniform time. $\square$

Corollary 4.5 (Near-perfect accuracy–time tradeoff). *For integers $n \geq 256$, $q \geq 2$ and rational parameters $0 < \kappa < 1$, $0 < \alpha < 5/8$, an instance with $\text{OPT} \geq 1 - \kappa$ admits a deterministically computed labeling with*

$$\text{err}(\ell) \leq \min \left\{ 1, C_{\text{GT}} \kappa \log(2n), C'' \sqrt{\frac{\kappa}{\alpha} \log \frac{2}{\alpha}} \right\}$$

in time $q^{O(n^\alpha)} \text{poly}(n, q, \text{input length})$. All constants are absolute, also for parameters varying with n .

Proof. Put $\beta = 8\alpha/5$, compute $l = \lceil \log_2 n \rceil$ and $k = \lceil \beta l \rceil$, and choose $R = \min\{n, \max\{256, 2^k\}\}$. Since $0 < \beta < 1$ and $\log_2 n \leq l < \log_2 n + 1$, we have $n^\beta \leq 2^k < 2^{1+\beta} n^\beta < 4n^\beta$. The cap at n preserves the lower bound, and the floor at 256 gives $n^\beta \leq R \leq 256n^\beta$. Thus $1 \leq v \leq 1/\beta$. Consequently

$$B_n \leq \frac{1 + \log(2/\beta)}{\beta} \leq \frac{3 \log(2/\beta)}{\beta} \leq \frac{3 \log(2/\alpha)}{\alpha}.$$

Theorem 4.4 applies, and $R^{5/8} \leq 256^{5/8} n^\alpha$. All parameter choices are finite integer or rational operations, with their encodings included in the input length. $\square$

Corollary 4.6 (Recovery without a supplied completeness promise). *Given only a rational weighted Unique Games instance with $n \geq 256$, $q \geq 2$, and an integer $256 \leq R \leq n$, a deterministic algorithm returns a labeling satisfying*

$$\text{err}(\ell) \leq \min\{1, C_{\text{GT}} \varepsilon_* \log(2n), C_{\text{pf}} \sqrt{\varepsilon_* B_n}\}, \quad \varepsilon_* = 1 - \text{OPT},$$

where $B_n = (\log n / \log R)(1 + \log(2 \log n / \log R))$, $C_{\text{GT}} = 600$, and C_{pf} is absolute. Its time is $q^{O(R^{5/8})} \text{poly}(n, q, \text{input length})$, with at most $O(1 + \log \log n)$ bounded runs of the algorithm of Theorem 4.4. Neither an upper bound on ε_ nor its value is an input.*

Proof. The finite geometric guessing construction, including termination under false guesses, is proved in Appendix B.1. $\square$

5 Actual-channel information and its limitations

The decomposition proof charges killing to a lower bound on the information of a split. We compute the full information and show that even this quantity cannot give a uniform bound $K/I = O(\eta/\log R)$ over the entire high-trace class. The same example passes the algorithm's low-rank stopping test: this is a limitation of an unrestricted local argument, not a lower bound on the algorithm's performance.

For a probability law b and $f \geq 0$, write $\text{Ent}_b(f) = \mathbb{E}_b[f \log f] - (\mathbb{E}_b f) \log(\mathbb{E}_b f)$, with $0 \log 0 = 0$. Let $b_{xy} = \mu(x)P(x, y)$ be a normalized symmetric endpoint law on a finite vertex set with positive marginal μ . For a finite nonnegative squared partition of unity $\sum_{i=1}^k \chi_i(x)^2 = 1$, define

$$w_i(x, y) = \chi_i(x)\chi_i(y), \quad w_{\dagger}(x, y) = 1 - \sum_i w_i(x, y), \quad p_z = \mathbb{E}_b w_z.$$

Cauchy–Schwarz makes these a probability channel from $E = (x, y)$ to the child or killed outcome $B \in \{1, \dots, k, \dagger\}$. Set $K = p_{\dagger}$ and $I = I(E; B)$. In the binary case we index the children by 0, 1. At a node of the construction, use the actual rounded multipliers passed to the children; the identities below do not require rationality. We write $D_{\text{KL}}(p||q) = \sum_{e: p(e) > 0} p(e) \log(p(e)/q(e))$ when q is positive on the support of p .

Lemma 5.1 (The exact split channel). *For every positive-probability outcome put $A_z = \{(x, y) : b_{xy}w_z(x, y) > 0\}$, $a_z = b(A_z)$, and $b_z(x, y) = b_{xy}w_z(x, y)/p_z$. Then*

$$K = \sum_i \mathcal{E}_P(\chi_i), \tag{8}$$

$$I = \sum_z \text{Ent} * b(w_z) = \sum *z : p_z > 0 p_z [\log(1/a_z) + D_{\text{KL}}(b_z || b(\cdot | A_z))]. \tag{9}$$

For the stereographic split $\chi_0 = (1 - r^2)/(1 + r^2)$, $\chi_1 = 2r/(1 + r^2)$, $0 \leq r \leq 1$, writing $D_{xy} = (1 + r_x^2)(1 + r_y^2)$ gives

$$(w_0, w_1, w_{\dagger}) = \frac{((1 - r_x^2)(1 - r_y^2), 4r_x r_y, 2(r_x - r_y)^2)}{D_{xy}}, \quad \mathcal{E}_P(r) \leq K \leq 4\mathcal{E}_P(r).$$

Proof. Summing $\mathcal{E}_P(\chi_i) = \mathbb{E}_{\mu} \chi_i^2 - \mathbb{E}_b w_i$ proves the first identity. Mutual information equals $\sum_{z: p_z > 0} p_z D_{\text{KL}}(b_z || b) = \sum_z \text{Ent}_b(w_z)$. On A_z , the law $b(\cdot | A_z)$ is b/a_z , giving the second identity by substitution. Substituting the stereographic multipliers gives the displayed probabilities; $1 \leq D_{xy} \leq 4$ and $\mathcal{E}_P(r) = \frac{1}{2} \mathbb{E}_b (r_x - r_y)^2$ give the energy comparison. $\square$

For each positive-mass outcome, its relative-entropy remainder vanishes if and only if $w_z = p_z/a_z$ b -almost surely on A_z . Thus all remainders vanish precisely when each such likelihood is constant on its support. This holds for indicator multipliers of a hard vertex partition: the child likelihoods indicate within-part pairs and the killed likelihood indicates between-part pairs. The remainders are therefore not uniformly positive.

Killing also carries information. For a nonzero soft restriction Φ of a half-lazy root law a , the notation of Section 2 gives

$$b\{x = y\} = \frac{\sum_x a_{xx} \Phi(x)^2}{Z_{\Phi}} \geq \frac{M_{\Phi}}{2Z_{\Phi}} \geq \frac{1}{2}.$$

Since $w_{\dagger}(x, x) = 0$, we have $a_{\dagger} \leq 1/2$ whenever $K > 0$. Hence (9) gives $I \geq p_1 \log(1/a_1) + K \log 2$ for a binary split, with the first term omitted if $p_1 = 0$ and the second interpreted as zero if $K = 0$. This is an aggregate diagonal-mass assertion, not pointwise half-laziness of the descendant walk.

Proposition 5.2 (A full-channel obstruction, and its rank screen). *Let $d \geq 2$ be an integer, let Q independently flip every coordinate of $\{0, 1\}^d$ with probability $1/(4d)$, and put $\rho = 1 - 1/(2d)$ and $P = 2Q - I$. This is a rational half-lazy walk with uniform stationary law. For $U \geq 2$, let*

$$T_U = \min\{t \in \mathbb{Z}_{\geq 0} : (1 + \rho^{2t})^d \leq 2^{d/U}\}.$$

Every nonnegative binary squared partition of unity with both child supports nonempty and proper satisfies

$$0 < K, \quad 0 < I \leq 6dK, \quad \frac{K}{I} \geq \frac{1}{6d} \geq \frac{\log U}{12T_U}. \quad (10)$$

For integer $U \geq 2$ and $d \geq 16U$, put

$$n = 2^d, \quad \nu = 1/n, \quad s = \lfloor d/U \rfloor, \quad R = 2^s, \quad T = T_U, \quad \eta = s/(32T).$$

Then

$$\mathrm{Tr}(Q^{2T}) > \sqrt{R}, \quad \frac{\log R}{\eta} \frac{K}{I} \geq \frac{8}{3}(\log 2) \log U, \quad \mathrm{rank} \mathbf{1}_{[0, \eta]}(I - P) < R^{5/8}. \quad (11)$$

These are rational algorithm parameters, with $256 \leq R \leq n$, $0 < \eta < 1/16$, and $\lfloor \log_2 R \rfloor / (32\eta) = T_U$, the heat time prescribed in Lemma 3.2. Moreover, with $u_h = \log(1/\nu)/\log R$, $U \leq u_h \leq 16U/15$ and $3072\eta u_h \leq 192/\log U$.

Proof. We give the entropy estimate, full-channel bound, and spectral computation separately. The cube entropy–energy inequality is classical [22, Exercises 10.23 and 10.26]; we include its proof in our normalization rather than use it as a black box.

Entropy and energy. For the uniform two-point law and $h = (a, b) \geq 0$, $\mathrm{Ent}(h^2) \leq (a - b)^2/2$. The case $a = b = 0$ is immediate. Otherwise homogeneity and, if necessary, interchanging a, b reduce to $a^2 = 1 + t$, $b^2 = 1 - t$, $0 \leq t \leq 1$. The derivative of the right side minus the left side is $t/\sqrt{1-t^2} - \mathrm{arctanh} t \geq 0$: its derivative is $(1-t^2)^{-3/2} - (1-t^2)^{-1} \geq 0$, and both differences vanish at zero. Continuity covers $t = 1$. For independent finite random variables X, Y and $f \geq 0$, entropy tensorizes because

$$\mathrm{Ent} *X, Yf = \mathbb{E}_Y \mathrm{Ent} Xf + \mathrm{Ent} Y(\mathbb{E}_X f) \leq \mathbb{E}_Y \mathrm{Ent} Xf + \mathbb{E}_X \mathrm{Ent} Yf.$$

The inequality uses convexity of entropy: for positive f , its second directional derivative is $\mathbb{E}(k^2/f) - (\mathbb{E}k)^2/\mathbb{E}f \geq 0$ by Cauchy–Schwarz; continuity covers zeros. Iteration and the two-point estimate give

$$\mathrm{Ent} \mu(h^2) \leq \frac{1}{2} \sum j = 1^d \mathbb{E} \mu(h(x) - h(x^{(j)}))^2 = d\mathcal{E}P * \mathrm{cube}(h),$$

where $x^{(j)}$ flips coordinate j of x and P_{cube} flips one uniformly selected coordinate. Bernoulli’s inequality gives $P(x, x^{(j)}) = (2d)^{-1}(1 - 1/(4d))^{d-1} \geq 3/(8d)$. Keeping just these nonnegative contributions in the Dirichlet form gives $\mathcal{E}_P(h) \geq (3/8)\mathcal{E}_{P_{\mathrm{cube}}}(h)$. Consequently

$$\mathrm{Ent}_\mu(h^2) \leq (8d/3)\mathcal{E}_P(h). \quad (12)$$

Full information. For every finite nonnegative squared partition of unity on any normalized symmetric endpoint law (not just the cube),

$$I \leq \sum_i \mathrm{Ent} * \mu(\chi_i^2) + K(1 + 2 \log(1/K)), \quad (13)$$

with the final term zero at $K = 0$. To prove it, for $0 \leq q \leq v$, $M = \mathbb{E}v$ and $d' = M - \mathbb{E}q$, write $h_{\text{bin}}(z) = -z \log z - (1-z) \log(1-z)$. When $M > 0$, the identity

$$\text{Ent}(q) + \text{Ent}(v-q) - \text{Ent}(v) = Mh * \text{bin}(d'/M) - \mathbb{E}[vh_{\text{bin}}(q/v)]$$

(with zero summands omitted), nonnegativity of entropy, and $h_{\text{bin}}(z) \leq z \log(e/z)$ give $\text{Ent}(q) \leq \text{Ent}(v) + d' \log(eM/d')$. If $M = 0$, both q and v vanish almost surely and the bound is immediate; if $d' = 0$, interpret its last term as zero. Apply this with $q_i = \chi_i(x)\chi_i(y)$ and $v_i = (\chi_i(x)^2 + \chi_i(y)^2)/2$. Convexity gives $\text{Ent}_b(v_i) \leq \text{Ent}_\mu(\chi_i^2)$, while $M_i = \mathbb{E}_b v_i = \mathbb{E}_\mu \chi_i^2$, $d_i = M_i - \mathbb{E}_b q_i = \mathcal{E}_P(\chi_i)$, $\sum_i d_i = K$, and $\sum_i M_i = 1$. Jensen's inequality gives $\sum_i d_i \log(eM_i/d_i) \leq K \log(e/K)$. Finally $0 \leq w_{\dagger} \leq 1$ implies $\text{Ent}_b(w_{\dagger}) \leq K \log(1/K)$, proving (13). Terms of zero mass are interpreted by continuity.

Walsh characters of degree j have Q -eigenvalue ρ^j , so $I - P$ has gap $2(1 - \rho) = 1/d$. As the two child supports are proper, there are distinct vertices x_0, x_1 where $z = (\chi_0, \chi_1)$ equals $e_0 = (1, 0)$ and $e_1 = (0, 1)$, respectively. They are distinct because $\|z(x)\|_2 = 1$ everywhere. For any vector v , $\|e_0 - v\|_2^2 + \|e_1 - v\|_2^2 = 1 + 2\|v - (e_0 + e_1)/2\|_2^2 \geq 1$. The spectral-gap inequality, applied to each coordinate of z , therefore gives

$$K \geq \frac{1}{d} \mathbb{E}_\mu \|z - \mathbb{E}_\mu z\|^2 \geq \frac{1}{d2^d},$$

where the last bound keeps just x_0, x_1 , each of mass 2^{-d} . Combining (12) and (13) gives

$$I \leq K \left(\frac{8d}{3} + 1 + 2d \log 2 + 2 \log d \right) \leq 6dK.$$

For the last inequality, $(1 + 2 \log d)/d$ decreases for $d \geq 2$, and $19/6 + 3 \log 2 < 6$. Positive loops give $p_1 > 0$, and proper support gives $a_1 < 1$, so (9) implies $I > 0$.

Time, trace, and rank. Bernoulli's inequality gives $Q(x, x) = (1 - 1/(4d))^d \geq 3/4$, so $P(x, x) \geq 1/2$. Off-diagonal entries of P are positive, its row sums are one, and it is symmetric and rational. This verifies the asserted half-lazy walk and uniform stationary law. Walsh expansion gives $\text{Tr}(Q^{2t}) = (1 + \rho^{2t})^d$ and

$$T_U = \left\lceil \frac{\log(1/(2^{1/U} - 1))}{-2 \log \rho} \right\rceil, \quad \frac{d}{2} \log U \leq T_U \leq 4d \log U.$$

Here $T_U \geq 1$ because the defining inequality fails at $t = 0$. For the time bounds use $(\log 2)/U \leq 2^{1/U} - 1 \leq 1/U$ and $1/d \leq -2 \log \rho \leq 2/d$. In particular the lower bound follows before taking the ceiling, while the upper bound follows from $T_U \leq d \log(U/\log 2) + 1 \leq 4d \log U$ for $d, U \geq 2$. These prove (10) and the middle inequality in (11), since $\log R/\eta = 32T_U \log 2$. Put $F_t = d \log(1 + \rho^{2t})$. Concavity and $\log(1 + 0) = 0$ give $F_T \geq \rho^2 F_{T-1}$. The minimality of T gives $F_{T-1} > d \log 2/U$, hence $F_T > \rho^2 d \log 2/U > \frac{1}{2} s \log 2$, proving the trace assertion. Here $s \geq 16$ and $\rho^2 \geq 9/16$. Also $U \leq d/s \leq 16U/15$ and $\eta u_h = d/(32T) \leq 1/(16 \log U)$. For integer U , the test defining T_U is equivalently $(1 + \rho^{2t})^U \leq 2$, an exact rational comparison, so T_U can be found by finite search. Since $\log_2 R = s$ and $32\eta = s/T$, the heat-time formula in Lemma 3.2 equals T exactly.

Finally let $m \in \{0, \dots, d\}$ be the largest degree with $2(1 - \rho^m) \leq \eta$. Since $\eta \leq 1/(16U \log U) < 1/16$, the defining inequality for m and $-\log(1 - z) \leq z/(1 - z)$ give

$$m \leq \frac{-\log(1 - \eta/2)}{-\log \rho} \leq \frac{d\eta}{1 - \eta/2} \leq \frac{2s}{31 \log U} < \frac{s}{15 \log U} =: a < d/2.$$

If $m = 0$, the rank is one. Otherwise the inclusive cutoff rank r obeys

$$r = \sum_{j=0}^m \binom{d}{j} \leq \exp(dh_{\text{bin}}(m/d)) \leq \exp(m \log(ed/m)) \leq \exp\left(\frac{s \log(16eU \log U)}{15 \log U}\right).$$

The first inequality follows by setting $z = m/(d - m) \leq 1$ in $\sum_{j \leq m} \binom{d}{j} \leq z^{-m}(1 + z)^d$; the last inequality uses monotonicity of $x \log(ed/x)$ for $x < d$ and $d/s \leq 16U/15$. The function $[\log(16e) + y + \log y]/(15y)$ decreases for $y \geq \log 2$, and its value at $\log 2$ is below $2/5$, since $e \log 2 < 2$. Hence $\log r < 2s/5 < (5/8)s \log 2$, proving the rank bound. $\square$

Taking integer U arbitrarily large and an integer $d \geq 16U$ makes $3072\eta u_h$ arbitrarily small, while the normalized full-channel ratio is bounded below by a positive constant times $\log U$, uniformly over the proper binary splits in the proposition. For example, $\log U \geq 6528$ suffices for the hypothesis $3072\eta u_h \leq 1/34$ of Theorem 3.1; the displayed spectral and channel inequalities themselves hold for every integer $U \geq 2$, $d \geq 16U$. Thus even within the smallness regime no absolute constant C can guarantee the existence of a proper binary split with $K \leq C(\eta/\log R)I$ on every high-trace node.

Nevertheless, these roots are connected and their inclusive cutoff ranks satisfy the stopping test $r^8 \leq R^5$ of Theorem 3.1, before any split. Stationary masses equal ν , so no mass lies below the cleanup threshold either. In particular, the construction has the single leaf $\Phi = 1$ and total loss $D = 0$ on these roots. The example therefore does not rule out a log-free local estimate *under rank-screen failure*, or a better charge restricted to reached nodes. It gives no lower bound on whole-tree loss, labeling error, or runtime. The Fourier and entropy–energy ingredients are classical; the limitation is their application to this particular channel and parameter regime.

A From a low-rank restriction to a partial assignment

We use one conditional-marginal construction and select among its roundings by their actual partial-assignment charge. Its ingredients are the singleton embedding and column-selection theorems of Guruswami–Sinop [4, Theorems 5.2 and 6.9]. Their local-fragment approach [5] avoids constructing an entire moment hierarchy. We record the fragment and its finite construction because a theorem about the final labeling alone does not supply the interface needed to transport a soft restriction back to its invocation root.

A.1 The finite conditional-marginal ensemble

Lemma A.1 (A conditional-marginal ensemble at low base rank). *Let a normalized rational weighted Unique Games instance have positive degree measure μ , alphabet $q \geq 2$, and symmetric normalized Laplacian H . Suppose*

$$r = \text{rank } \mathbf{1}_{[0,\eta]}(H), \quad 0 < 128\kappa_0 \leq \eta \leq 1/4.$$

If its optimum is at least $1 - \kappa_0$, a deterministic algorithm returns at most q^{2r} rational simplex fields p^σ with positive rational weights w_σ summing to one, such that

$$\mathbb{E}_\sigma \hat{\mathcal{E}}(p^\sigma) \leq \frac{65}{64}\kappa_0, \quad \mathbb{E}_\sigma \sum_u \mu(u)(1 - \|p_u^\sigma\|^2) \leq \frac{5\kappa_0}{\eta}. \quad (14)$$

Its running time is $q^{O(r)} \text{poly}(n, q, b)$, where b is the binary input length, including η and κ_0 . Without the optimum promise, it returns an ensemble passing the displayed rational checks or declares failure in the same time. Parallel records and arbitrary permutation loops are allowed.

Proof. If $2r \geq n$, enumerate every labeling and retain a minimum-error one. This costs at most q^{2r} times polynomial work. If its error exceeds $65\kappa_0/64$, declare failure; otherwise return its one-hot field as a singleton ensemble of zero impurity. Hence assume $2r < n$.

The required moment fragment. Its singleton indicator vectors $X_{u,a}$ satisfy $\sum_a X_{u,a} = X_\emptyset$, $\|X_\emptyset\|^2 = 1$, and orthogonality within every fiber. Its objective is

$$F = \frac{1}{2} \sum_e a_e \sum_a \|X_{u,a} - X_{v,\pi_e(a)}\|^2 \leq b_0 := 65\kappa_0/64.$$

We construct a seed set S , $|S| \leq 2r$, and compatible joint indicators for $S \cup \{u\}$, for every u . If P_S is projection onto the seeded singleton span, the construction ensures

$$\sum_u \mu(u) \sum_a \|(I - P_S)X_{u,a}\|^2 \leq 5\kappa_0/\eta. \quad (15)$$

Seed selection from singleton inner products. The embedding in [4, Theorem 6.9] supplies unit vectors Y_u with

$$\|Y_u - Y_v\|^2 \leq 2 \sum_a \|X_{u,a} - X_{v,\pi(a)}\|^2, \quad \sum_a \|(I - P_S)X_{u,a}\|^2 \leq \|(I - Q_S)Y_u\|^2,$$

where Q_S projects onto the span of $(Y_v)_{v \in S}$. This embedding uses within-fiber orthogonality, not entrywise positivity of cross-fiber inner products. Indeed, for $A = \|x\| > 0$, $B = \|y\| > 0$ and $c = \langle x, y \rangle$, the matched-pair estimate follows from

$$A^2 + B^2 - 4c + 2c^2/(AB) = (A - B)^2 + 2AB(1 - c/(AB))^2 \geq 0.$$

For the Gram matrix G of the columns $\sqrt{\mu(u)}Y_u$, $\text{Tr}(GH) \leq 2F$, and therefore $\sum_{i>r} \lambda_i^\downarrow(G) \leq 2F/\eta$. The column-selection theorem [4, Theorem 5.2], with $2r$ columns, gives residual at most twice this spectral tail.

Here is a rational implementation retaining slack. Put $\delta = \kappa_0/(16n\eta)$ and compute a rational positive definite G_+ with $G \preceq G_+ \preceq G + 2\delta I$. Its entries use

$$G_{uv} = \sqrt{\mu(u)\mu(v)} \sum_{a,b} \frac{\langle X_{u,a}, X_{v,b} \rangle^2}{\sqrt{\|X_{u,a}\|^2 \|X_{v,b}\|^2}},$$

where a zero-denominator term is zero. Terms with denominator below $\tau = \delta/(16nq^2)$ can be omitted: positive semidefiniteness bounds each such term by τ . Their sum in any entry is at most $q^2\tau = \delta/(16n)$. Rational square-root bisection on the remaining terms produces a symmetric rational $\tilde{G}$ with entrywise error at most $\delta/(4n)$. Set $G_+ = \tilde{G} + \delta I$. The operator-norm error of $\tilde{G}$ is at most $\delta/4$, so $G + 3\delta I/4 \preceq G_+ \preceq G + 5\delta I/4$. All required precision lengths are polynomial in the rational input length and $\log(1/\delta)$.

Here are the exact conditional-expectation computations for selecting $2r$ columns. For a positive definite matrix D , put $e_j(D) = \sum_{|T|=j} \det D[T, T]$, with $e_0(D) = 1$; these are the coefficients of $\det(I + tD)$. Choosing k columns with weights proportional to their principal determinants gives expected Schur residual $(k+1)e_{k+1}(D)/e_k(D)$. Indeed each residual diagonal is a bordered determinant divided by the selected principal determinant, so each determinant of order $k+1$ is counted $k+1$ times. If D is the Schur complement after the already fixed choices and k more columns are needed, write D/i for the Schur complement of its diagonal pivot at i . The conditional expected final residual and its averaging weights are

$$E_i = \frac{ke_k(D/i)}{e_{k-1}(D/i)}, \quad p_i = \frac{D_{ii}e_{k-1}(D/i)}{ke_k(D)}.$$

They satisfy $\sum_i p_i = 1$ and $\sum_i p_i E_i = (k+1)e_{k+1}(D)/e_k(D)$ by the same determinant counting. Choose an index minimizing E_i , replace D by D/i , and decrement k until no choices remain. Characteristic-polynomial coefficients and Schur complements have polynomial rational bit cost; no family of n^{2r} subsets is enumerated. For eigenvalues $\lambda_1 \geq \dots \geq \lambda_n > 0$, each product of $k+1$ eigenvalues contains at least $k+1-r$ tail indices. Thus $(k+1-r)e_{k+1} \leq (\sum_{i>r} \lambda_i)e_k$. At $k=2r$, the initial expected residual is less than twice the tail beyond rank r . Loewner monotonicity of Schur residuals now gives

$$\sum_u \mu(u) \sum_a \|(I - P_S)X_{u,a}\|^2 \leq 4F/\eta + 4n\delta \leq \frac{69}{16} \frac{\kappa_0}{\eta} < 5\kappa_0/\eta.$$

This seed procedure reads only the singleton inner products.

Compatible local moments, without an $n^{O(r)}$ hierarchy. Use linearly independent centered indicator functions at each vertex, $z_{u,a} = \mathbf{1}_{\ell(u)=a} - 1/q$, $a < q$; eliminate the last indicator by the sum-to-one identity and reduce repeated-vertex products by

$$z_a^2 = (1 - 2/q)z_a + (q-1)/q^2, \quad z_a z_b = -(z_a + z_b)/q - 1/q^2 \quad (a \neq b).$$

The base moments are the monomials on at most two vertices; their nonconstant number is $d = n(q-1) + \binom{n}{2}(q-1)^2$. For a seed set of size s , index both rows and columns of the symmetric matrix M_S by the distinct reduced monomials supported in $S \cup \{u\}$ for some u , including the constant monomial. Its entry at two row indices is the formal expectation of their reduced product. Use one scalar moment variable for each distinct nonconstant monomial occurring in these products; all occurrences of the same monomial share that variable. There are $m_S \leq (1 + n(q-1))q^s$ rows and $N_S = O(n^2 q^{s+2})$ nonconstant moment coordinates. Let

$$K_S = \{y : M_S(y) \succeq 0, F(y) \leq b_0, -2 \leq y_A \leq 2\}.$$

The constant moment is fixed and all indicator identities have already been eliminated. Rational inertia gives a strong separator for this spectrahedron in time polynomial in its explicit size.

The following inradii justify finite local-fragment optimization. Mix a promised witness distribution with uniform independent labels, using uniform mass $\sigma = \kappa_0/256$. Its objective is at most $\kappa_0 + \sigma < b_0$. For each base monomial, the product of the appropriate functions $\psi_{u,a} = q(\mathbf{1}_{\ell(u)=a} - \mathbf{1}_{\ell(u)=q})$ is its dual under uniform measure and has absolute value at most q^2 . Thus a base perturbation of infinity norm at most

$$\rho_* := \frac{\kappa_0}{1024dq^2}$$

is realized by adding the corresponding dual linear combination to the mixture's density relative to uniform measure. Its absolute size is at most $dq^2\rho_* = \sigma/4$, and its mean is zero. The resulting distribution stays nonnegative and has objective less than b_0 . Consequently $C_* := \bigcap_{|S| \leq 2r} \text{proj}_{\text{base}} K_S$ contains a Euclidean ρ_* -ball. This body is used only in the proof and is never enumerated.

Each K_S also contains a local ball of explicitly known radius; its center need not be known to the algorithm. Use the moments of the same witness–uniform mixture as its center. Under uniform labels, different vertex supports of centered monomials are orthogonal, and the one-vertex covariance has minimum eigenvalue q^{-2} . The uniform row Gram matrix therefore has minimum eigenvalue at least $q^{-2(s+1)}$, so the mixture's Gram matrix has minimum eigenvalue at least $\sigma q^{-2(s+1)}$. A coordinate perturbation of Euclidean norm a changes M_S in operator norm by at most $m_S 2^{s+1}a$,

because reduced products have coefficient one-norm at most 2^{s+1} . The objective has coefficient one-norm at most $10q^2$. It follows that K_S contains a ball of radius

$$\rho_S = \min \left\{ \frac{1}{4}, \frac{\kappa_0}{4096q^2}, \frac{\kappa_0 q^{-2(s+1)}}{1024m_S 2^{s+1}} \right\}.$$

Indeed the last radius bound permits a Gram perturbation of at most one quarter of this minimum eigenvalue. The objective increases by at most $10q^2\rho_S \leq 10\kappa_0/4096$, smaller than its slack at the mixture, and the moment box has slack at least one at the center. In particular $\log(1/\rho_S) = O(s \log q + \log n + \log(1/\kappa_0))$.

Run a shallow-cut ellipsoid in the d base coordinates, starting with a ball containing $[-2, 2]^d$. At a base point x , first test the box, objective, and base PSD conditions, the latter using $M_{\emptyset}(x)$ with constant and centered-singleton row indices. A violated condition supplies a separating cut. If all pass, select $S(x)$ by the preceding seed procedure and apply Lemma A.2 below to $K_{S(x)}$, with fixed base coordinates x and tolerance $\epsilon = \rho_*/(100d)$. An extension ends the algorithm. Otherwise the returned inequality $\langle c, z - x \rangle \leq \epsilon$, $\|c\|_{\infty} = 1$, holds for every $z \in C_*$. Every current ellipsoid contains its ρ_* -ball; its directional half-width is at least ρ_* , so this is a valid shallow cut of relative depth at most $1/(100d)$. After $O(d^2 \log(d/\rho_*))$ unsuccessful cuts the volume would be smaller than that ball, a contradiction. The algorithm therefore returns a fragment with exactly the base moments that selected its seeds. Rounded rational ellipsoid updates with fixed precision are used throughout [10, Section 3.3]. Keep violated box cuts exact. For all remaining calls the center x lies in the box; round each normalized cut direction to infinity error at most $\epsilon/(16d)$ and add the corresponding $4d$ times that error to its offset; this preserves validity and changes the relative depth by less than a factor two. This also prevents successive oracle calls from inflating the outer coefficient length. All dimensions are $q^{O(r)}$ poly(n, q), and the logarithms of the inverse radii and precisions are polynomial in the stated binary inputs. Fixed budgets and exact output checks give the asserted failure behavior without the promise.

Rounding the fragment. Let $X_S(\sigma)$ be its mutually orthogonal seed indicators, $w_{\sigma} = \|X_S(\sigma)\|^2$, and omit zero weights. They sum to $X_{\emptyset}$, so the positive weights sum to one. Set

$$p_u^{\sigma}(a) = \frac{\langle X_S(\sigma), X_{u,a} \rangle}{w_{\sigma}}.$$

The seed-singleton indicator is a local row combination; its idempotence and PSD show that its moment is nonnegative. Exact indicator identities show that each p_u^{σ} is a simplex vector. The projection Π_S onto the seed-indicator span contains the seeded singleton span, and orthogonal projection gives

$$\mathbb{E}_{\sigma} \hat{\mathcal{E}}(p^{\sigma}) \leq F, \quad \mathbb{E}_{\sigma} \sum_u \mu(u)(1 - \|p_u^{\sigma}\|^2) = \sum_u \mu(u) \sum_a \|(I - \Pi_S)X_{u,a}\|^2 \leq 5\kappa_0/\eta.$$

These are precisely (14); retain all positive-weight seeds, rather than imposing separate seedwise energy and purity tests. The simplex, normalization, and two mean inequalities can be checked exactly. Small positive weights affect bit length, not repetition counts. A label-changing loop contributes the nonnegative unary objective $a_e \sum_{a:\pi_e(a) \neq a} \|X_{u,a}\|^2$; hence none of the preceding arguments discard such constraints. $\square$

Lemma A.2 (Fixed-coordinate extension or a shallow cut). *Let $K \subset [-2, 2]^N$ be a rational strongly separable convex body containing a ball of known rational radius $0 < \rho \leq 1/4$. Given integers*

$1 \leq d \leq N$, rational fixed coordinates $x \in [-2, 2]^d$, and rational $0 < \epsilon < 1$, one can return either a rational $y \in K$ with $y_{\text{base}} = x$, or rational $c \in \mathbb{R}^d$ with

$$\|c\|_\infty = 1, \quad \langle c, z_{\text{base}} - x \rangle \leq \epsilon \quad (z \in K).$$

The time is polynomial in the dimensions, oracle and input lengths, $\log(1/\rho)$, and $\log(1/\epsilon)$. With these budgets imposed, failure is permitted when the inradius promise is false.

The cut alternative is allowed even if the fiber is nonempty. The lemma does not assert an exact decision procedure for feasibility of an arbitrary lower-dimensional spectrahedral slice.

Proof. This is the inradius-aware restricted-support construction underlying [5]; the short proof specifies its geometric margin. If $N = d$, use the strong oracle. Otherwise put $\delta = \epsilon\rho/(128\lceil\sqrt{dN}\rceil)$ and $\beta = (\epsilon\delta/(64d))^2$. Run a rational ellipsoid on the affine fiber, retaining all full-space cuts and the enclosing box. A feasible point ends the algorithm. A violated inequality with zero free part is already a separator in the base coordinates. Otherwise finite volume reduction returns a polytope $K \subset P \subset [-2, 2]^N$ whose fixed-coordinate fiber has volume less than a free-coordinate ball of radius δ .

Write $P = \{z : C_i z \leq b_i\}$. Choose rational $\|C_i\|_2 \leq h_i \leq 2\|C_i\|_2$ and set $P' = \{z : C_i z \leq b_i - 2\delta h_i\}$. It contains a $\rho/2$ -ball and lies inside the radius- 2δ erosion of P . Minimize $\|z_{\text{base}} - x\|^2$ over P' to additive accuracy β , obtaining a rational feasible point $\hat{y}$. This requires only ordinary bounded convex quadratic optimization with the known inradius [10, Section 3.3]. More explicitly, let $f(z) = \|z_{\text{base}} - x\|^2$ and $0 < \gamma \leq 1$. If y_* is a minimizer and $B(w, \rho/2) \subset P'$, the points $(1-t)y_* + tz$, $z \in B(w, \rho/2)$, with $t = \gamma/(32d)$, form a ball of radius $\gamma\rho/(64d)$. Convexity and $f(z) \leq 16d$ throughout $P' \subset [-2, 2]^N$ place this entire ball in the sublevel $f \leq f(y_*) + \gamma/2$. Ellipsoid feasibility of such sublevels and binary search therefore have bit-polynomial budgets in $\log(1/\rho) + \log(1/\beta)$; no unknown center is computed.

Let $v = \hat{y}_{\text{base}} - x$ and let v_* be the base displacement of a minimizer. The thin-fiber certificate and the 2δ erosion imply $\|v\| > \delta$; otherwise the erosion ball would cut the fiber in a ball of radius greater than δ . Optimality gives $\|v - v_*\| \leq \sqrt{\beta}$. If w is a ρ -ball center in P , then for every $z \in P$, $z_0 = (1 - 4\delta/\rho)z + (4\delta/\rho)w$ lies in P' and $\|z - z_0\| \leq 16\delta\sqrt{N}/\rho$. First-order optimality gives $\langle v_*, (z_0)_{\text{base}} - x \rangle \geq \|v_*\|^2 \geq 0$. Consequently $c = -v/\|v\|_\infty$ satisfies

$$\langle c, z_{\text{base}} - x \rangle \leq \frac{4d\sqrt{\beta}}{\delta} + \frac{16\delta\sqrt{dN}}{\rho} < \epsilon.$$

Certify this inequality over the rational polytope P by rational linear programming before returning it. Check any proposed extension with the original strong oracle as well. A failed budget, failed check, or zero normal to be normalized results in failure, not an unchecked output; these exits cannot occur under the promised conditions. Thus failed inradius promises cannot install invalid cuts. Each ellipsoid call uses bounded rational updates and its predetermined volume budget. All stated inverse precisions occur logarithmically, proving the running-time assertion. $\square$

A.2 Ensemble selection with separate disagreement and amplitude costs

All quantities in this subsection use one normalized symmetric directed invocation root. Its scalar law is half-lazy. For a nonempty partial labeling on U' , write B for its directed internal violated mass and $C_{\text{out}} = \sum_{u \in U', v \notin U'} a_{uv}$, and define its charge

$$\Gamma = \frac{B + 2C_{\text{out}}}{\mu(U')}.$$

Let $g_u = s_u e_{\ell(u)}$ with $s_u \geq 0$; labels at zero amplitudes are arbitrary. Define the root quantities

$$\begin{aligned} M(g) &= \sum_u \mu(u) s_u^2, & A(g) &= \mathcal{E}_P(s), \\ J(g) &= \sum_{e:u \rightarrow v} a_e s_u s_v \mathbf{1}_{\ell(v) \neq \pi_e(\ell(u))}, & \Omega(g) &= \sum_u \sum_{v \neq u} a_{uv} s_u^2, \\ \mathcal{I}(g) &= \sum_{e:u \rightarrow v} a_e \min\{s_u^2, s_v^2\} \mathbf{1}_{\ell(v) \neq \pi_e(\ell(u))}, & \mathcal{V}(g) &= \frac{1}{2} \sum_{u,v} a_{uv} |s_u^2 - s_v^2|. \end{aligned}$$

Lemma A.3 (Disagreement-sensitive ensemble coarea). *For each such field,*

$$\hat{\mathcal{E}}(g) = A(g) + J(g), \quad 0 \leq A(g) \leq \Omega(g) \leq M(g)/2, \quad \mathcal{I}(g) \leq J(g), \quad \mathcal{V}(g)^2 \leq A(g)(2\Omega(g) - A(g)). \quad (16)$$

For a finite ensemble with positive weights summing to one, let bars denote weighted means and assume $\bar{M} > 0$. Threshold each field on $U_{\sigma,t} = \{u : s_u^2 > t\}$ for $t \geq 0$. Its least nonempty seed-threshold charge satisfies

$$\Gamma_* \leq \frac{\bar{\mathcal{I}} + 2\bar{\mathcal{V}}}{\bar{M}} \leq \frac{\bar{J}}{\bar{M}} + \frac{2\sqrt{\bar{A}(2\bar{\Omega} - \bar{A})}}{\bar{M}} \leq j + 2\sqrt{a(1-a)}, \quad (17)$$

where $a = \bar{A}/\bar{M} \in [0, 1/2]$ and $j = \bar{J}/\bar{M}$. For rational data the minimum is computable exactly with at most n thresholds per nonzero field and polynomial work in the ensemble encoding.

Proof. On every directed record, with permutation pullback understood,

$$\|s_u e_{\ell(u)} - T_e s_v e_{\ell(v)}\|^2 = (s_u - s_v)^2 + 2s_u s_v \mathbf{1}_{\ell(v) \neq \pi_e(\ell(u))}.$$

Summing proves $\hat{\mathcal{E}} = A + J$. Also $A = \Omega - \sum_{u \neq v} a_{uv} s_u s_v \leq \Omega$, and half-laziness gives $\Omega \leq M/2$. The inequality $\min\{s_u^2, s_v^2\} \leq s_u s_v$ gives $\mathcal{I} \leq J$. After discarding scalar loops, Cauchy–Schwarz gives

$$\mathcal{V}^2 \leq \left(\frac{1}{2} \sum_{u \neq v} a_{uv} (s_u - s_v)^2 \right) \left(\frac{1}{2} \sum_{u \neq v} a_{uv} (s_u + s_v)^2 \right) = A(2\Omega - A).$$

For one field, integrating over $t \in [0, \infty)$, coarea gives exactly $\int \mu(U_t) dt = M$, $\int B_t dt = \mathcal{I}$, and $\int C_{\text{out},t} dt = \mathcal{V}$: internal records survive an interval of length $\min\{s_u^2, s_v^2\}$, and a directed boundary record contributes $(s_u^2 - s_v^2)_+$. Symmetry pairs opposite boundary orientations. Loops contribute only to the internal term. Average $\Gamma_{\sigma,t}$ with density $w_\sigma \mu(U_{\sigma,t})/\bar{M}$ to prove the first bound. Another Cauchy–Schwarz inequality gives $\bar{\mathcal{V}} \leq \sqrt{\bar{A}(2\bar{\Omega} - \bar{A})}$, proving the rest. For finite selection sort the distinct positive rational s_u^2 and test every interval midpoint, including the interval from zero to the smallest level. All nonempty thresholds occur and all charges are rational. Zero fields are omitted. No square roots or random draws from possibly tiny seed masses are used in selection. $\square$

For a nonzero root multiplier $\chi \geq 0$, use

$$M_\chi = \sum_u \mu(u) \chi(u)^2, \quad Z_\chi = \sum_e a_e \chi(u) \chi(v) > 0, \quad E_\chi = M_\chi - Z_\chi, \quad r_\chi = E_\chi / Z_\chi.$$

The normalized restriction has record weights $a_e \chi(u) \chi(v) / Z_\chi$ and degree law μ_χ . Write $\mathcal{E}_\chi(\cdot)$ and $\hat{\mathcal{E}}_\chi(\cdot)$ for its scalar and constraint energies, respectively; the scalar $E_\chi = M_\chi - Z_\chi$ continues to denote the root multiplier energy. Scalar half-laziness gives $Z_\chi \mu_\chi(u) \geq \mu(u) \chi(u)^2 / 2$ and $0 < Z_\chi \leq M_\chi \leq 2Z_\chi$, so $0 \leq r_\chi \leq 1$.

Theorem A.4 (Transported ensemble and the sharper leaf selector). *Suppose a finite rational simplex ensemble on the restriction, with positive weights summing to one, has mean energy at most F and mean impurity at most ρ . Let $W(p)_a = (p_a - \max_{b \neq a} p_b)_+$ and set $g^\sigma = \chi W(p^\sigma)$, extending by zero outside the support. If $D_0 = M_\chi - 4Z_\chi \rho > 0$, then*

$$\overline{M} \geq D_0, \quad 0 \leq \overline{J} \leq Z_\chi F, \quad \overline{A} \leq \left(\sqrt{2Z_\chi F - 2\overline{J}} + \sqrt{E_\chi} \right)^2. \quad (18)$$

Writing $d_0 = D_0/Z_\chi$ and $z = \overline{J}/Z_\chi \in [0, F]$, the least actual seed-threshold charge obeys

$$\Gamma_* \leq \frac{z}{d_0} + \frac{2}{\sqrt{d_0}} \left(\sqrt{2(F - z)} + \sqrt{r_\chi} \right). \quad (19)$$

For the ensemble of Lemma A.1, in particular,

$$\Gamma_*^2 \leq \frac{520}{27} \kappa_0 + \frac{256}{27} r_\chi. \quad (20)$$

For that constructed ensemble there are at most nq^{2r} seed-threshold pairs, with rational polynomial-time selection per pair. The selected seed need not obey separate energy or impurity thresholds.

Proof. The permutation-equivariant winner-gap map has at most one positive coordinate and is $\sqrt{2}$ -Lipschitz: on each ordering region its only nonzero row is the winning coordinate minus the runner-up. Continuity and subdivision of line segments prove the global assertion. If $a \geq b$ are the two largest coordinates of a simplex vector, then $\|p\|^2 \leq a^2 + b(1 - a)$ and $a + b \leq 1$, giving $2\|p\|^2 - (a - b)^2 \leq a^2 + 2b - b^2 \leq 1$. Therefore

$$\|W(p)\| \leq 1, \quad 1 - \|W(p)\|^2 \leq 2(1 - \|p\|^2).$$

Degree domination consequently yields $M(\chi W(p)) \geq M_\chi - 4Z_\chi \sum_u \mu_\chi(u)(1 - \|p_u\|^2)$; average to obtain $\overline{M} \geq D_0$.

We also retain how the winner map allocates its energy. If h, k are the two output gaps and i, j their winners, then

$$(h - k)^2 + 4hk\mathbf{1}_{i \neq j} \leq 2\|p - q\|^2.$$

For a zero gap or coincident winners this follows from Lipschitzness. Otherwise $h + k \leq (p_i - q_i) - (p_j - q_j) \leq \sqrt{2}\|p - q\|$, whose square is the desired left side. Apply this recordwise, including permutation loops. With $d_u = \|W(p_u)\|$, it gives $\mathcal{E}_\chi(d) + 2J_\chi(W(p)) \leq 2\hat{\mathcal{E}}_\chi(p)$. Disagreement transports exactly: $J(\chi W(p)) = Z_\chi J_\chi(W(p))$. Thus $\overline{J} \leq Z_\chi F$ and the mean leaf amplitude energy is at most $2F - 2\overline{J}/Z_\chi$.

For $a \geq b \geq 0$ and $x, y \in [0, 1]$, write $ax - by = b(x - y) + (a - b)x$; reverse the roles for $b > a$. The triangle inequality over weighted edge spaces gives

$$\sqrt{\mathcal{E}_P(\chi d)} \leq \sqrt{Z_\chi \mathcal{E}_\chi(d)} + \sqrt{E_\chi}.$$

Apply the same triangle inequality in the weighted direct sum over seeds. It proves the amplitude estimate in (18). Lemma A.3 gives $\Gamma_* \leq \overline{J}/\overline{M} + 2\sqrt{\overline{A}/\overline{M}}$; substituting the preceding bounds proves (19).

For the constructed ensemble, $F = 65\kappa_0/64$, $\rho = 5\kappa_0/\eta$, and $d_0 = M_\chi/Z_\chi - 20\kappa_0/\eta \geq 27/32$. Put $f = F/d_0$ and $x = z/d_0 \in [0, f]$. Since $\kappa_0 \leq 1/512$, we have $f < 2$, and

$$2\sqrt{2f} - 2\sqrt{2(f - x)} = \frac{2\sqrt{2}x}{\sqrt{f} + \sqrt{f - x}} \geq x.$$

The zero case follows by continuity. Hence $\Gamma_* \leq 2(\sqrt{2F} + \sqrt{r_\chi})/\sqrt{d_0}$. Squaring and using $(a+b)^2 \leq 2a^2 + 2b^2$ gives $\Gamma_*^2 \leq (16F + 8r_\chi)/d_0$, proving (20). Exact finite selection is supplied by Lemma A.3; no additional moment extension is required. $\square$

Corollary A.5 (The leaf decoder used by the outer algorithm). *Suppose a visited restriction has base cutoff rank at most $R^{5/8}$ at $0 < \eta = 1024\bar{\kappa} \leq 1/4$, optimum at least $1 - 8\bar{\kappa}$, and $r_\chi \leq 8D$. A deterministic algorithm returns a nonempty partial assignment satisfying*

$$\Gamma^2 \leq \frac{65}{432}\eta + \frac{2048}{27}D \quad (21)$$

at cost $q^{O(R^{5/8})}$ times polynomial work in the binary input.

Proof. Apply Lemma A.1 and Theorem A.4 with $\kappa_0 = 8\bar{\kappa}$. Then $128\kappa_0 = \eta$, and substitution in (20) gives the displayed constants. $\square$

The total tree loss D is used only for existence; selection and acceptance use the actual rational partial-label charge.

A.3 Direct multiplier thresholds and a rank-certified menu

There is a complementary candidate: round labels throughout the support of χ and threshold only χ^2 . Here we use the invocation root's stronger property that its *identity-loop* mass at u is at least $\mu(u)/2$. No half-laziness of the normalized leaf is assumed. Put

$$m_e = a_e \min\{\chi(u)^2, \chi(v)^2\}, \quad V_\chi = \frac{1}{2} \sum_{u,v} a_{uv} |\chi(u)^2 - \chi(v)^2|.$$

For a simplex field on the support, define the rational quantity

$$Q_\chi(p) = \sum_e m_e b_e(p), \quad b_e(p) = \begin{cases} 1 - \sum_a p_u(a) p_v(\pi_e(a)), & u \neq v, \\ \sum_{a: \pi_e(a) \neq a} p_u(a), & u = v. \end{cases} \quad (22)$$

Every min-overlap sum below omits terms with $m_e = 0$, and vertex sums involving marginals are restricted to the support of χ . Thus no labels or probabilities outside that support are required.

Theorem A.6 (Direct multiplier-threshold selection). *For any finite rational simplex ensemble with positive weights summing to one, mean leaf energy at most F and mean impurity at most ρ , deterministic selection produces a nonempty partial labeling with*

$$\Gamma_{\text{dir}} \leq T_\chi := \frac{\min_\sigma Q_\chi(p^\sigma) + 2V_\chi}{M_\chi} \leq \frac{Z_\chi(F + c\rho) + 2V_\chi}{M_\chi}, \quad (23)$$

$$\Gamma_{\text{dir}} \leq \frac{F + c\rho + 2\sqrt{r_\chi}}{1 + r_\chi}, \quad c = \begin{cases} 1/2, & \text{all root loops are identities,} \\ 1, & \text{otherwise.} \end{cases} \quad (24)$$

There is no purity-denominator condition. If $\chi = \mathbf{1}_C$, replace $2\sqrt{r_\chi}$ by $2r_\chi$. The cost is polynomial in the full ensemble encoding, with at most n thresholds per seed and no additional moment construction.

Proof. For a labeling L of the support put $I_\chi(L) = \sum_e m_e \mathbf{1}_{L(v) \neq \pi_e(L(u))}$. Independent labels at distinct vertices give $\mathbb{E} I_\chi(L) = Q_\chi(p)$. A loop uses one random label, which explains the unary, not bilinear, second line of (22). For each seed, fix vertices successively to labels of positive probability minimizing the remaining conditional expectation. The averaged value equals the previous value, so

the final labeling satisfies $I_\chi(L) \leq Q_\chi(p)$. Each step evaluates sums involving at most two remaining probabilities, with polynomial rational bit cost; no inverse-probability sampling is used.

We prove the impurity coefficient, including label-changing loops. Let $C_u = \text{Diag}(p_u) - p_u p_u^\top \succeq 0$ and $\delta_u = \text{Tr } C_u = 1 - \|p_u\|^2$. Write $c_u = \sum_{e:u \rightarrow v, v \neq u} m_e$ and $A_m = \frac{1}{2} \sum_e m_e \|p_u - T_e p_v\|^2 \leq Z_\chi \mathcal{E}_\chi(p)$. For a non-loop record, its expected violation equals its energy contribution plus $(\delta_u + \delta_v)/2$. For a loop, the difference is $\text{Tr}((I - T_e)C_u)$. Thus symmetry gives the exact identity

$$Q_\chi(p) = A_m + \sum_u c_u \delta_u + \sum_{e:u \rightarrow u} m_e \text{Tr}((I - T_e)C_u).$$

Only the symmetric part of T_e contributes; its operator norm is at most one, so the loop trace is between zero and $2\delta_u$, and vanishes on identity loops. Denote the two root loop masses by l_u^{id} and l_u^{non} . Identity half-laziness gives $l_u^{\text{id}} \geq l_u^{\text{non}} + \sum_{v \neq u} a_{uv}$, whence

$$c_u + 2\chi(u)^2 l_u^{\text{non}} \leq \chi(u)^2 (l_u^{\text{id}} + l_u^{\text{non}}) \leq Z_\chi \mu_\chi(u).$$

If all loops are identities, the stronger two bounds $c_u \leq \chi(u)^2 l_u^{\text{id}}$ and $c_u \leq Z_\chi \mu_\chi(u) - \chi(u)^2 l_u^{\text{id}}$ instead give $2c_u \leq Z_\chi \mu_\chi(u)$. Consequently, in both cases $Q_\chi(p) \leq Z_\chi(\mathcal{E}_\chi(p) + c \sum_u \mu_\chi(u) \delta_u)$.

The coarea identities of Lemma A.3, with amplitudes $s = \chi$ and labels L , give a multiplier threshold of charge at most $(I_\chi(L) + 2V_\chi)/M_\chi$. Its scalar bound also gives $V_\chi^2 \leq E_\chi(M_\chi - E_\chi) = E_\chi Z_\chi$. For $\chi = \mathbf{1}_C$, direct computation instead gives $V_\chi = E_\chi$. Enumerate all nonempty multiplier thresholds for every derandomized seed, including the interval below the smallest positive squared multiplier, and retain the least actual charge. This proves the first bound in (23). The minimum of Q_χ is at most its ensemble mean, proving the second. Use $M_\chi = Z_\chi(1 + r_\chi)$ for the final envelopes. All threshold and score comparisons are rational, and their number is polynomial. $\square$

For the ensemble of Lemma A.1, substitute $F = 65\kappa_0/64$ and $\rho = 5\kappa_0/\eta$. For an integer cap $r_{\max}$ at least the inclusive η -cutoff rank, one can additionally test the rational cutoffs $\eta, 2\eta, 4\eta, \dots$ below $1/4$, together with $1/4$, keeping only those ζ whose inclusive leaf rank is at most $r_{\max}$. Rational inertia of the generalized matrix $L_\chi - \zeta D_\chi$, counting nonpositive eigenvalues, certifies that rank. Here $D_\chi = \text{diag}(\mu_\chi)$ and $L_\chi = D_\chi - a^\chi$ are the normalized restriction's degree and unnormalized Laplacian matrices. Every retained cutoff satisfies $128\kappa_0 \leq \eta \leq \zeta \leq 1/4$, so the same marginal construction applies with unchanged energy bound and impurity at most $5\kappa_0/\zeta$. The menu has $O(1 + \log(1/\eta))$ entries, polynomial in the binary input; its total cost is $q^{O(r_{\max})} \text{poly}(n, q, b)$. No larger cutoff is asserted to pass on an arbitrary leaf.

Retain the winner-gap candidates at the original cutoff when adjoining these direct candidates. For use in the outer algorithm set $r_{\max} = \max\{r \in \mathbb{Z}_{\geq 0} : r^8 \leq R^5\}$ and apply this cap to every retained cutoff. Keep the original acceptance thresholds and stopping rules. The old guaranteed candidate is still present, and taking the least actual charge preserves all stated approximation and search guarantees. The menu makes only polynomially many additional bounded decoder calls per inspected leaf, each taking $q^{O(R^{5/8})} \text{poly}(n, q, b)$ time; this preserves the stated global time class, not necessarily the running time at a particular lower-rank leaf. In particular, the direct certificate may improve an observed charge but does not authorize skipping the terminal-mass charge or any invocation test.

B The classical fallback and promise-free recovery

The uniform running-time statement uses the logarithmic approximation of Gupta–Talwar [6]. We give the weighted rational implementation and its deterministic selection step; neither the

approximation factor nor the metric-tree method is claimed as new. The construction below does not use the supplied completeness parameter in its computation. The final subsection uses this implementation to remove the supplied completeness promise from the uniform recovery theorem.

Lemma B.1 (Finite weighted tree embedding). *Given a positive rational metric d on $v \geq 2$ terminals and rational weights $w_{ab} \geq 0$ on unordered pairs, a deterministic polynomial-time algorithm constructs a rational 2-hierarchically separated tree metric D such that*

$$D(a, b) \geq d(a, b), \quad \sum_{a < b} w_{ab} D(a, b) \leq 32 \log(2v) \sum_{a < b} w_{ab} d(a, b).$$

Here outgoing edges at each internal node have equal lengths, and edge lengths decrease by a factor two down the tree. Polynomial time is measured in the full rational input length.

Proof. This is the FRT construction [9, Sections 2.1–2.3 and 2.5], with its finite conditional-expectation implementation made explicit. Scale d by a power of two σ so that the minimum positive distance of $\tilde{d} = \sigma d$ lies in $(1, 2]$, and put $h = \lceil \log_2 \text{diam}(\tilde{d}) \rceil \geq 1$. Both choices use rational comparisons; h is bounded by the input bit length. For $1 < \alpha < 2$ and a permutation of all terminals, start from one root cluster at level h . At levels $i = h - 1, \dots, 0$, partition each parent by assigning every terminal in it to the first center at distance less than $r_i = \alpha 2^{i-1}$ in $\tilde{d}$. Centers range over all terminals, also those outside the parent. A child is indexed by its parent and center. Join a level- i child to its parent by an edge of length $2^{i+1}/\sigma$, keeping a node even when its cluster equals its parent. Level zero consists of singletons, so the tree has at most $1 + vh$ nodes.

If (a, b) first separates at level $i < h - 1$, its parent lies in a ball of radius $r_{i+1} < 2^{i+1}$, so $\tilde{d}(a, b) < 2^{i+2}$, whereas its tree path contains two edges of length $2^{i+1}/\sigma$. For separation immediately below the root use $\text{diam}(\tilde{d}) \leq 2^h$ instead. This proves domination. Summing the geometric lengths gives $D(a, b) < 8 \cdot 2^i/\sigma$. Let C_i indicate that the first center whose radius- r_i ball contains either endpoint contains exactly one. At first separation $C_i = 1$, whence

$$D(a, b) \leq (8/\sigma) \sum_{i=0}^{h-1} 2^i C_i. \quad (25)$$

For a fixed pair let $a_z = \min\{\tilde{d}(a, z), \tilde{d}(b, z)\}$, $b_z = \max\{\tilde{d}(a, z), \tilde{d}(b, z)\}$, $A(r) = \{z : a_z < r\}$, and $B(r) = \{z : a_z < r \leq b_z\}$. For a uniform permutation the probability of $C_i = 1$ at radius r is $|B(r)|/|A(r)|$. Order centers by increasing a_z , breaking ties arbitrarily. The j th center, when in $B(r)$, has at least j centers in $A(r)$; also $b_z - a_z \leq \tilde{d}(a, b)$. Hence

$$\int_0^\infty \frac{|B(r)|}{|A(r)|} dr \leq \sum_{j=1}^v \frac{b_j - a_j}{j} \leq H_v \tilde{d}(a, b), \quad H_v = \sum_{j=1}^v \frac{1}{j}.$$

Averaging also over uniform $\alpha \in (1, 2)$ and substituting $r = \alpha 2^{i-1}$ gives $\mathbb{E} \sum_i 2^i C_i \leq 2H_v \tilde{d}(a, b)$. Thus (25) bounds the expected weighted cost by $16H_v \sum w_{ab} d(a, b) \leq 32 \log(2v) \sum w_{ab} d(a, b)$.

For derandomization use the rational estimator

$$\Psi(\alpha, p) = (8/\sigma) \sum_{a < b} w_{ab} \sum_i 2^i \Pr[C_i = 1 \mid \alpha, p],$$

where p is a permutation prefix. If p has encountered $A(r_i)$, its first such center fixes C_i ; otherwise the conditional probability is exactly $|B(r_i)|/|A(r_i)|$, since none of these centers has been used. All center sets change only at rational breakpoints $\alpha = 2^{1-i} \tilde{d}(a, z)$ in $(1, 2)$. Enumerate the $O(v^2 h)$

breakpoints, adjoin the endpoints 1, 2, and sort and remove duplicates. Minimize $\Psi(\alpha, \emptyset)$ over rational interval midpoints. Since it is constant on each open interval, the minimum is no larger than its continuous average. This includes the single interval (1, 2) when there are no interior breakpoints. Then choose each next center to minimize the conditional estimator. Its average over remaining centers is the current value, so it cannot increase. The final permutation and (25) prove the claim. All enumerations are polynomial; distances, interval midpoints, and dyadic lengths have polynomial bit length. Conditional probabilities have denominators at most v , so a common denominator for every estimator is a product of input weight denominators, the denominator of $1/\sigma$, and $\text{lcm}(1, \dots, v)$, with dyadic factors of $O(h)$ bits. Thus comparisons also have polynomial bit cost. $\square$

Lemma B.2 (Deterministic weighted logarithmic approximation). *For a weighted rational Unique Games instance on $n \geq 1$ vertices, there is a deterministic algorithm, polynomial in n, q and the instance bit length, returning a labeling of error at most*

$$\min\{1, 600\kappa \log(2n)\} \quad \text{whenever} \quad \text{OPT} \geq 1 - \kappa.$$

Parallel constraints, nonidentity loops, disconnected graphs, and arbitrarily small positive rational weights are allowed.

Proof. Discard zero-weight records. If none remain, any labeling suffices; otherwise normalize their weights to sum to one. Each record $e = (u, v)$ has an orientation and permutation π_e from u to v . Use variables $x(u, a), d(e, a) \in [0, 1]$ with

$$\sum_a x(u, a) = 1, \quad d(e, a) \geq |x(u, a) - x(v, \pi_e(a))|, \quad L_{\text{LP}} = \frac{1}{2} \sum_e w_e \sum_a d(e, a).$$

The absolute value represents two linear constraints. In the undirected label lift, the edge from (u, a) to $(v, \pi_e(a))$ has length $d(e, a)$. For each lifted root $s = (r, a)$ add potentials $t_s(z) \in [0, 1]$ satisfying

$$t_s(s) = 0, \quad t_s(r, b) \geq x(r, a) \ (b \neq a), \quad t_s(j) \leq t_s(i) + d(e, c)$$

in both orientations of every lifted edge $i-j$. This is a polynomial-size rational LP: it has $O(n^2q^2 + mq)$ variables and $O(nmq^2 + n^2q^2)$ constraints for m records. Minimize L_{LP} by exact rational linear programming [10, Theorem 6.4.12]. Every conflict walk W from (r, a) to (r, b) , $b \neq a$, obeys

$$x(r, a) \leq \sum_{(e, c) \in W} d(e, c). \quad (26)$$

Every integral labeling is feasible: use one-hot marginals, their absolute differences for d , and truncated shortest-path distances from each lifted root for t_s , taking distance one for unreachable states. A walk from a true label to a false label crosses a length-one edge. Its LP objective is exactly its error, including loops. Thus the rational optimum has $L_{\text{LP}} \leq \kappa$ and polynomial encoding length.

If $L_{\text{LP}} = 0$, every lift length is zero because all remaining weights are positive. In each non-loop connected component choose a root label with positive marginal and propagate it along a spanning tree. The marginal of each propagated label stays positive because the edge marginals agree under transport. Any violated record, including a loop, would form, with the two root paths, a conflict walk moving the positive root label. This contradicts (26). Isolated vertices use their own positive-marginal label. This returns a satisfying labeling.

Assume henceforth $L_{\text{LP}} > 0$. Put $\ell_e = \sum_a d(e, a)$ and, for non-loop records, $\ell'_e = \ell_e + L_{\text{LP}}/(10n)$. Shortest-path completion gives a positive rational metric on each non-loop component

and $\sum_{e \text{ non-loop}} w_e \ell'_e \leq 2L_{\text{LP}} + L_{\text{LP}}/(10n)$. Apply Lemma B.1 componentwise, adding weights of parallel records for the embedding only. Eliminate the tree's Steiner vertices as follows. Represent each nested cluster by its least terminal in a fixed order, and contract edges with equal representatives. Every representative's fiber is connected and contains its terminal leaf, so the quotient is a tree on terminals. Give each surviving edge the original metric distance between its representatives. For an old edge of length L from a parent to a child, each representative is a descendant terminal of the parent and is at tree distance less than $2L$ from it, by geometric decrease. Their original metric distance is thus less than $4L$, by domination. The image of any old terminal-to-terminal path is the unique quotient-tree path after contracted edges are removed. Summing the surviving edge bounds gives $D_T \leq 4D$, while the triangle inequality for the original metric gives $D_T \geq d$. Represent each new tree edge by an original shortest path for ℓ' , with its composed permutation. Its unperturbed path length is at most its tree length. Singleton components need no embedding.

For a tree edge with transport σ , write $p_a = x(u, a)$, $q_b = x(v, b)$, $m_a = \min\{p_a, q_{\sigma(a)}\}$, $r_a = p_a - m_a$, $s_b = q_b - m_{\sigma^{-1}(b)}$, and $t = \sum_a r_a = \sum_b s_b$. The rational coupling

$$Y(a, b) = \mathbf{1}_{b=\sigma(a)} m_a + r_a s_b / t$$

uses only its first term at $t = 0$. It has marginals p, q and mismatch probability $t = \frac{1}{2} \sum_a |p_a - q_{\sigma(a)}|$, at most half the represented path length by telescoping. Indeed $r_a s_{\sigma(a)} = 0$. Root each tree, orient the couplings away from the root, and set $K_{uv}(a, b) = Y(a, b)/p_a$ when $p_a > 0$, using any rational probability row when $p_a = 0$. Together with the root marginal these tables define a rational Markov law on labelings, with marginal $x(v, \cdot)$ at every vertex; take the product law across components.

For a non-loop original record $e = (u, v)$, expand the tree path into its representing paths and close it by the reverse of e . For each starting label moved by this closed walk, (26) bounds its marginal by the lifted walk length. Summing over moved labels gives at most $D_T(u, v) + \ell_e$, since every prefix transport is a permutation. If the label is not moved and all tree edges on the path agree with their transports, the record is satisfied. The union bound on tree mismatches therefore gives $\Pr[e \text{ violated}] \leq \ell_e + 2D_T(u, v)$. For a loop, (26) gives $x(v, a) \leq d(e, a)$ whenever $\pi_e(a) \neq a$, so the violation probability is at most ℓ_e . Consequently

$$\mathbb{E}[\text{err}] \leq 2L_{\text{LP}} + 256 \log(2n)(2L_{\text{LP}} + L_{\text{LP}}/(10n)) \leq 600L_{\text{LP}} \log(2n).$$

For the last inequality, divide by $L_{\text{LP}} \log(2n)$ and use $\log(2n) \geq \log 2 > 1/2$ and $n \geq 1$: the coefficient is at most $4 + 512 + 128/5 < 600$.

Finally derandomize this law exactly. For a partial assignment F , let $\psi_v(b)$ indicate compatibility with its assignment at v . For a nonroot vertex v with parent u compute the upward message

$$M_v(a) = \sum_b K_{uv}(a, b) \psi_v(b) \prod_{z \text{ child of } v} M_z(b).$$

For each root r , sum $Z_r(F) = \sum_b x(r, b) \psi_r(b) \prod_{z \text{ child of } r} M_z(b)$. Their product is $Z(F) = \Pr[X \text{ extends } F]$. Inconsistent clamps have probability zero. For $Z(F) > 0$ the conditional expected error is

$$\frac{\sum_e w_e [Z(F) - \sum_a Z(F, X_u = a, X_v = \pi_e(a))]}{Z(F)}.$$

For each successive vertex, choose a positive-probability label minimizing this expression. The average over those labels is the current value, so it cannot increase. After all choices the actual error is at most the initial expectation, and it is trivially at most one.

Each probability evaluation uses $O(nq^2)$ rational operations, and only polynomially many evaluations are required. A product of all root and transition-table denominators is a common

denominator of all complete-label probabilities, so exact messages and conditional comparisons have polynomial bit length even for tiny probabilities. The optimal LP solution, shortest paths, and metric aspect ratio have polynomial rational encoding length; Lemma B.1 therefore also runs in polynomial bit time. No numerical factor $1/\kappa$ or $1/L_{LP}$ is used. $\square$

B.1 Removing the supplied completeness promise

Proof of Corollary 4.6. We retain the corollary's hypotheses $n \geq 256$ and $q \geq 2$. Discard zero-weight records before any constraint check. First test satisfiability exactly. In every positive-weight non-loop connected component, try each label of one root, propagate it along a spanning tree, and check every record, including loops and parallel constraints. The component is satisfiable precisely when a root label passes all checks. Isolated vertices use the same test on their loops; zero-degree vertices are arbitrary. This is polynomial work. If all components pass, return the resulting satisfying labeling. Zero total weight is also handled this way. Otherwise $\varepsilon_* > 0$.

Run the algorithm of Lemma B.2. It needs no completeness parameter: its perturbation is determined by the rational LP optimum. The proof of that lemma gives an output ℓ_0 with

$$\varepsilon_* \leq a := \text{err}(\ell_0) \leq \min\{1, C_{GT}\varepsilon_* \log(2n)\}, \quad C_{GT} = 600.$$

In particular $a > 0$, and it is computed exactly from rational record weights. Set

$$l = \lceil \log_2(2n) \rceil, \quad J = \lceil \log_2(C_{GT}l) \rceil, \quad \kappa_j = a/2^j \quad (0 \leq j \leq J).$$

Retain only guesses satisfying $1/n \leq \kappa_j < 1$. Run the deterministic algorithm of Theorem 4.4 with each retained guess, with the exhaustion convention specified below. Return the labeling of least actual rational error among these outputs and ℓ_0 . No decision compares unknown optimum values or expressions involving real logarithms.

Bounded runs under false guesses. In a deterministic localization invocation, enumerate its prescribed finite set of paths and bounded decoder calls. If none supplies a candidate passing the actual rational charge test, assign the remaining vertices arbitrarily and terminate that guessed run. Successful invocations each remove a nonempty set, so there are at most n of them before completion or this exhaustion exit. The graph construction, rank tests, support reduction, rational entropy estimates, path count, and bounded decoder failure behavior do not require a completeness witness. Only the proof that some candidate passes uses the promise. Thus an incorrect guess cannot force an unbounded search, and it is never assumed to give a good labeling.

These bounded runs have the claimed uniform time even without a witness. Theorem 4.4 invokes localization only when $R > \lceil \log_2 n \rceil^4$; on this branch the unconditional cap $h_U \leq 2 \log n$ absorbs the deterministic path exponent into $O(R^{5/8} \log q)$. Every retained guess has $1/\kappa_j \leq n$, so numerical inverse-completeness work is polynomial in n . On the other branch only the polynomial-time fallback is run. When a guess is valid, the original good-leaf proof guarantees a passing candidate before exhaustion, so the convention does not change Theorem 4.4's guarantee.

A sufficiently accurate valid guess, or the fallback. If $1/n \leq \varepsilon_* < 1/2$, then

$$1 \leq a/\varepsilon_* \leq C_{GT} \log(2n) \leq C_{GT}l.$$

The integer $j = \lfloor \log_2(a/\varepsilon_*) \rfloor$, used only in this existence proof, lies in $[0, J]$ and satisfies

$$\varepsilon_* \leq \kappa_j < 2\varepsilon_* < 1.$$

It is retained and valid. Writing C' for the constant in Theorem 4.4, its output has error at most $\sqrt{2C'}\sqrt{\varepsilon_*B_n}$. If $0 < \varepsilon_* < 1/n$, the fallback instead satisfies

$$C_{\text{GT}}\varepsilon_* \log(2n) \leq C_{\text{GT}}\sqrt{\varepsilon_*} \frac{\log(2n)}{\sqrt{n}} \leq C_{\text{GT}}\sqrt{\varepsilon_*B_n},$$

using $\log(2n) \leq \sqrt{n}$ and $B_n \geq 1$. For $\varepsilon_* \geq 1/2$, the bound follows from error at most one and $2\sqrt{\varepsilon_*B_n} \geq 1$. The case $\varepsilon_* = 0$ was already handled by the exact test. Thus $C_{\text{pf}} = \max\{\sqrt{2C'}, C_{\text{GT}}, 2\}$ suffices; retaining ℓ_0 gives the linear branch simultaneously.

Finally $J = O(1 + \log \log n)$. All guesses have encoding length polynomial in the original input length, since a is an exactly evaluated rational error and division by 2^j adds only j bits. The bounded run costs, number of calls, and exact final comparisons prove the stated running time without an input completeness bound. $\square$

C Whole-tree loss, guarded policies, and certified flat partitions

This optional refinement measures the loss of a complete invocation, not the ratio of one split. Here a policy specifies every node decision and tie rule before its validation samples are drawn. It defines a finite binary tree of nonnegative rational local multipliers with $\chi_0^2 + \chi_1^2 = 1$ and nonempty proper child supports. Each child law is the normalized soft restriction of its parent, and each branch draw uses the child-or-death probabilities of Theorem 3.1. Thus every path has depth at most $n - 1$. Admissibility additionally requires the stated leaf-rank, loss-cap, and uniform polynomial path-access guarantees; these are hypotheses to be verified for each policy, not consequences of its pilot estimates. For a fixed policy, the telescoping calculation of Theorem 3.1 gives

$$D = \sum_{v \text{ internal}} Z_v K_v = 1 - \sum_{b \text{ leaf}} Z_b = \Pr\{\text{an exact path is killed}\}. \quad (27)$$

Pilot paths stop only at death or a terminal leaf; cleanup deaths count. Conditioning on survival or counting decoder failures would estimate a different quantity. These pilots never run a leaf decoder.

C.1 Estimating and comparing complete policies

Lemma C.1 (Bounded rational loss estimation). *Fix a deterministic binary policy with the access guarantees of Theorem 3.1 and a rational cap $0 \leq D \leq d \leq 1/2$. For rational $0 < \epsilon, \delta < 1$, put*

$$b = \lceil \log_2(4/\delta) \rceil, \quad N = \lceil 16(d + \epsilon/2)b/\epsilon^2 \rceil.$$

There is a bounded procedure using at most N paths which, with probability at least $1 - \delta$, returns rational endpoints satisfying

$$0 \leq L_D \leq D \leq U_D \leq d, \quad U_D - L_D \leq \epsilon.$$

It can instead declare failure, included in the exceptional probability. Every nonfailure output, even an exceptional one, has $0 \leq L_D \leq U_D \leq d$ and width at most ϵ . The bit cost is $N \text{poly}(n, b_{\text{in}}, 1/\eta, \log(N/\delta))$. For $d = 0$ return $[0, 0]$ without sampling.

Proof. On independent ideal exact paths, death indicators Y_j are Bernoulli with mean D . For $|\lambda| \leq 1$,

$$\log \mathbb{E} e^{\lambda(Y_j - D)} \leq D(e^\lambda - 1 - \lambda) \leq D\lambda^2.$$

Here the first inequality is $\log(1+x) \leq x$ and the second follows from the exponential series. Thus either tail of $\hat{D} = N^{-1} \sum_j Y_j$ at distance s is at most $\exp(-N\lambda s + ND\lambda^2)$ for $0 < \lambda \leq 1$. Take $s = \epsilon/2$ and $\lambda = s/(2(d+s))$. The exponent is at most $-Ns^2/(4(d+s)) \leq -b$, so $|\hat{D} - D| \leq \epsilon/2$ except with probability $2e^{-b} \leq \delta/2$. Output

$$U_D = \min\{d, \hat{D} + \epsilon/2\}, \quad L_D = \min\{U_D, \max\{0, \hat{D} - \epsilon/2\}\}.$$

These intervals have the asserted unconditional formatting properties and contain D on the stated deviation event.

For bounded rational sampling, clear the three branch-probability denominators to an integer B . Draw $\lceil \log_2 B \rceil$ fair bits, reject the represented integer when it is at least B , and otherwise use the three integer intervals of lengths Bp_0, Bp_1, BK . Each trial accepts with probability greater than $1/2$ (and probability one when B is a power of two). Cap each draw at $k = \lceil \log_2(2Nn/\delta) \rceil$ trials, accepting a valid final trial, and declare pilot failure if any draw exhausts its cap. At most $N(n-1)$ draws occur. Coupling to ideal draws using the same bits bounds the probability of any discrepancy by $Nn2^{-k} \leq \delta/2$. We do not condition on avoiding failure. Combining the two exceptional probabilities proves the confidence claim. The path access bound, polynomial denominator lengths, and rational power comparisons prove the bit bound. $\square$

Proposition C.2 (Guarded comparison of complete policies). *At one fixed root, let $m \geq 1$ admissible deterministic policies with losses $D_1, \dots, D_m$ be fixed, with policy 1 the baseline. Each must independently have the same certified analytic loss cap, rank, partition-of-unity, and polynomial path-access guarantees. Apply Lemma C.1 with scale ϵ and allowance δ/m to each policy. Exactly known losses may instead use singleton intervals. If any pilot declares failure, keep the baseline and its analytic upper cap. Otherwise let j minimize U_j , and choose j only when $U_j \leq L_1$; if this fails, retain policy 1. With probability at least $1 - \delta$,*

$$D_{\text{sel}} \leq D_1, \quad D_{\text{sel}} \leq U_{\text{sel}} \leq \min\{D_{\text{sel}} + \epsilon, \min_k D_k + 2\epsilon\}. \quad (28)$$

The cost is the sum of the pilot costs, not an enumeration of all policies.

Proof. With probability at least $1 - \delta$, no pilot declares failure and all intervals contain their losses. On this event $U_j \leq \min_k D_k + \epsilon$. If the guard accepts, $D_j \leq U_j \leq L_1 \leq D_1$. Otherwise $U_1 \leq L_1 + \epsilon < U_j + \epsilon \leq \min_k D_k + 2\epsilon$. In both cases the selected interval contains its loss and has width at most ϵ , so $U_{\text{sel}} \leq D_{\text{sel}} + \epsilon$. The union bound requires only the validity of each pilot, not independence between different policies. $\square$

C.2 Exactly certified flat partitions

At an admitted residual root of Theorem 4.2, retain its parameters and the rational budget of Lemma 4.3, and define

$$\bar{\kappa} = \kappa/t, \quad \eta = 1024\bar{\kappa}, \quad d_U = \min\{1/2, 3 \cdot 2^{17}\eta\bar{B}_U + \kappa/50\}, \quad A_U = 2^{38}\bar{\kappa}\bar{B}_U. \quad (29)$$

The component-first construction has $D \leq d_U$. Admission gives $\eta\bar{B}_U \leq 2^{-24}$ and $\kappa \leq 2^{-34}$, so $d_U < 1/32$. An explicit partition $\mathcal{P} = \{C_1, \dots, C_s\}$ into nonempty sets proposes leaf laws and a complete-tree loss

$$z_j = \sum_{x,y \in C_j} a_{xy}, \quad a_{xy}^j = a_{xy}/z_j \quad (x, y \in C_j), \quad D_{\mathcal{P}} = 1 - \sum_j z_j.$$

Accept this proposal only if exact rational inertia verifies $\text{rank } \mathbf{1}_{[0,\eta]}(I - P_j) \leq R^{5/8}$ for every part and the rational test $D_{\mathcal{P}} \leq d_U$ passes. Explicitly, for each part form $A_j = (a_{xy})_{x,y \in C_j}$ and $D_j = \text{diag}(\sum_{y \in C_j} a_{xy})_{x \in C_j}$. The diagonal is positive by the root loops. Count the nonpositive eigenvalues of $(1 - \eta)D_j - A_j$ by rational inertia and test $r_j^8 \leq R^5$. Division by z_j does not change this count.

Proposition C.3 (Certified flat candidates). *Every passing partition is admissible at this invocation. Its loss is known exactly, and its at most n leaves can all be decoded within $nq^{O(R^{5/8})}$ times polynomial bit cost. At least one resulting candidate has actual root charge*

$$\Gamma^2 \leq \min\{A_U, a_{\text{rd}}\eta + b_{\text{rd}}D_{\mathcal{P}}\}, \quad a_{\text{rd}} = 65/432, \quad b_{\text{rd}} = 2048/27. \quad (30)$$

Proof. Successive hard cuts isolating the parts give a binary squared partition of unity with leaves $\mathbf{1}_{C_j}$ and leaf probabilities $z_j > 0$; positivity follows from the root loops. Its loss is the crossing ordered mass $D_{\mathcal{P}}$, independently of isolation order. The identities in Theorem 3.1 and the proof of Lemma 3.4 apply directly: they require these global identities and the loss cap, not the heat-selection rule. Hence some leaf is good. The checked rank and Corollary A.5 give $\Gamma^2 \leq a_{\text{rd}}\eta + b_{\text{rd}}D_{\mathcal{P}}$. This also passes the baseline because, using $\kappa \leq \eta/1024$ and $\bar{B}_U \geq 1$,

$$a_{\text{rd}}\eta + b_{\text{rd}}d_U \leq \left(2^{28}/9 + 65/432 + 1/675\right) \eta \bar{B}_U < 2^{28} \eta \bar{B}_U = A_U.$$

All leaf decoders are bounded, including on nongood leaves; enumerate them and minimize actual charge. All admission tests are rational. $\square$

Such a flat candidate has a certified invocation-level loss bound; it need not obey the sharper standalone entropy coefficient. It may be included in Proposition C.2 with its exact singleton interval. No claim is made that every proposed partition passes the tests.

C.3 The end-to-end actual-loss guarantee

Theorem C.4 (An actual-loss-sensitive complete-labeling guarantee). *Under Theorem 4.2, at each admitted root fix a finite portfolio containing the component-first baseline. Every other implicit policy must independently satisfy that theorem’s conditional-entropy loss cap and the access and rank guarantees above; include only flat proposals passing Proposition C.3. Suppose the portfolio size is polynomially bounded, its proposals are constructed in polynomial time (or supplied as part of the input), and node access has a uniform polynomial bound. There is a bounded randomized implementation with the same running-time class as Theorem 4.2 which, with probability at least $1 - \xi$, retains its H_* error guarantee and additionally satisfies*

$$\text{err}(\ell) \leq \min \left\{ 1, 2 \left[\sum_i \Delta_i \sqrt{\min\{A_i, a_{\text{rd}}\eta_i + b_{\text{rd}}V_i\}} + t_f \right] \right\}. \quad (31)$$

Here i ranges over accepted invocations on the realized residual trajectory, $t_i = \mu^0(U_i)$, $\Delta_i = t_i - t_{i+1}$, $\eta_i = 1024\kappa/t_i$, A_i is (29), and

$$V_i = \min \left\{ D_{i,\text{sel}} + \eta_i, \min_k D_{i,k} + 2\eta_i, \min_{\mathcal{P}} D_{i,\mathcal{P}} \right\}.$$

The last minimum is over that root’s passing flat proposals and is $+\infty$ if there are none. Moreover $D_{i,\text{sel}} \leq D_{i,1}$ simultaneously. Loss comparisons are at the same realized root, not between different adaptive trajectories. The original tail and budget stops, and terminal mass t_f , are retained.

Proof. At each root freeze its policies and use Proposition C.2 with $\epsilon = \eta$ and total allowance $\delta = \xi/(2n)$. On declared pilot failure use the baseline and $U = d_U$; otherwise use its selected upper endpoint U . Decode all passing flat-partition leaves and retain their least-charge baseline-passing candidate. Write D_{flat} for the least exact flat loss; if there is no flat candidate, omit its threshold from the following minimum. On fresh randomness try at most $J = \lceil \log_2(4n/\xi) \rceil$ paths of the selected policy. At each leaf run the bounded decoder and retain the least actual charge among candidates passing $\Gamma^2 \leq A_U$. Accept early when

$$\Gamma^2 \leq \min\{A_U, a_{\text{rd}}\eta + b_{\text{rd}}U, a_{\text{rd}}\eta + b_{\text{rd}}D_{\text{flat}}\}. \quad (32)$$

The same test is made on retained flat candidates. If no tightened acceptance occurs, use the best retained baseline candidate; if none exists, declare production-search failure and complete arbitrarily as in the original bounded implementation. Killed paths and leaf-decoder failures only end their attempts.

Every accepted candidate passes A_U , regardless of pilot accuracy. Therefore the deterministic charge, residual-envelope, and terminal-stop arguments of Theorem 4.2 remain valid on every run without production-search failure, giving the same H_* bound. This uses the immutable residual inequalities, not equality of the old and new trajectories. For implicit candidates the required cap is the component-conditional cap; an arbitrary total-entropy policy is not silently assigned this smaller cap. Flat candidates instead have the separate exact certificate above.

Condition on the entire history before each validation. The root and its portfolio are then fixed, so the conditional union bound over at most n roots makes all pilots nonfailing, all intervals covering, and (28) valid except with probability $\xi/2$. For any fixed current pilot history with valid intervals, every good leaf of the selected policy gives a candidate with $\Gamma^2 \leq \min\{A_U, a_{\text{rd}}\eta + b_{\text{rd}}D_{\text{sel}}\}$, by Lemma 3.4 and Corollary A.5. If the flat threshold in (32) is smallest, its enumerated candidates already pass. Otherwise every good sampled leaf passes the tightened test, since $D_{\text{sel}} \leq U$. Thus J ideal attempts fail to give tightened acceptance with probability at most $2^{-J} \leq \xi/(4n)$. Without any pilot accuracy assumption the same argument with only A_U guarantees a baseline-passing candidate.

Cap each production branch draw at $\lceil \log_2(4n^2J/\xi) \rceil$ rejection trials, treating cap failure as a failed attempt. There are at most n^2J draws. Coupling to fresh ideal paths bounds any cap discrepancy by $\xi/4$, and the ideal search failures over all roots cost at most another $\xi/4$. Together with validation this gives total exceptional probability at most ξ . This argument conditions on history before the current samples, not on successful rejection draws or an optional stopping event.

On its complement every acceptance is tightened. The two upper bounds on U in (28), and the exact flat threshold, give the expression in (31). Charge accounting yields $\text{err} \leq \min\{1, 2(\sum_i \Delta_i \Gamma_i + t_f)\}$; the factor two undoes half-lazification. This proves the claimed bound.

At one root, every implicit policy needs $O((d_U + \eta)\eta^{-2} \log(nm/\xi)) = O(\bar{B}_U \eta^{-1} \log(nm/\xi))$ pilot paths. Here $\kappa \leq \eta/1024$, $\bar{B}_U \geq 1$, and $1/\eta \leq 1/(1024\kappa)$. Their costs are polynomial in the parameters already allowed in Theorem 4.2; they require no leaf decoders. The at most polynomially many flat leaves and nJ production leaves contribute only polynomially many bounded $q^{O(R^{5/8})}$ calls. All counts, means, tests and probability tables have polynomial bit length. This proves the same bounded running-time class. $\square$

For a single implicit policy without flat proposals, the new bound uses $(a_{\text{rd}} + b_{\text{rd}})\eta_i + b_{\text{rd}}D_i$. In particular it implies

$$\text{err}(\ell) \leq \min \left\{ 1, 128\sqrt{(a_{\text{rd}} + b_{\text{rd}})\kappa} + 2\sqrt{b_{\text{rd}} \sum_i \Delta_i D_i + 2t_f} \right\}.$$

Indeed $\sum_i \Delta_i / \sqrt{t_i} \leq \int_0^1 t^{-1/2} dt = 2$, and Cauchy–Schwarz gives $\sum_i \Delta_i \sqrt{D_i} \leq \sqrt{\sum_i \Delta_i D_i}$. This is a trajectory-sensitive consequence, not a claim that the displayed losses are uniformly small. The exactly recorded actual charges always certify $2(\sum_i \Delta_i \Gamma_i + t_f)$ on successful runs; confidence is needed to relate pilot bounds to true losses and to obtain efficient tightened acceptance. Guarding complete policies does not assert that greedily decreasing a local K/I decreases whole-tree loss. Policies modified after viewing validation data require fresh validation. These pilot-enhanced guarantees are randomized; deterministic exhaustive path search alone does not make their confidence statements deterministic.

Use of AI: The author prepared a two-page draft (without using AI) on treating spectral decomposition as an information-budgeted process. The author then fed this draft to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) and prompted them to apply it to improve the best known algorithms for Unique Games, work out the details of the proofs and identify missing gaps. The author fixed those missing gaps and iteratively verified and expanded the proofs using AI tools. The author is solely responsible for the correctness of the proofs.

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